Distributed Systems

Summary

Authors

Katharina Tschirky

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1. Week 1 - Scaling

1.1. Distributed Systems Motivation - Scaling

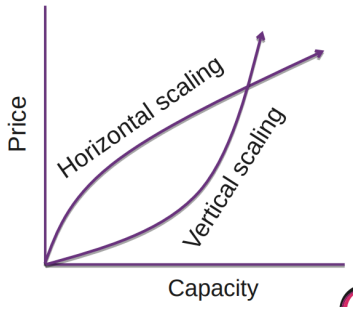
- Vertical (scale up), more memory, faster CPU
  - Single Core
    - Amdahl's Law: Your total speedup is limited by the part of the work that cannot be parallelized
    - S = 1 / ((1 - P) + P/N)
    - S = Speedup, P = parallelizable fraction (0–1), N = number of parallel units
    - Example: P = 95% -> even with infinite processors, the speedup can never be more than 20x
    - Thus the sequential part is always the bottleneck e.g in real word DB locks limit throughput regardless of server count
- Horizontal (scale out), more machines

Machine Learning:

- Current trend: scale horizontally
- ML with vertical scaling is not (yet) feasible

Economics:

- Initially scaling vertically is cheaper until HW is maxed out
- Currently: RAM prices increasing, fast servers



Horizontal Scaling

+ Lower cost with massive scale
+ Easier to add fault-tolerance
+ Higher availability
- Adaption of software required
- More complex system, more components involved

Vertical Scaling

+ Lower cost with small scale
+ No adaption of software required
+ Less complexity
- HW limits for scaling
- Risk of HW failure causing outage
- More difficult to add fault-tolerance

1.2. Vertical Scaling Performance

Notable laws:

- Moore's Law: nr. of transistors doubles

every 2 years

- Nielsen's Law: a high-end user's

connection speed grows by 50% per year

- Kryder'S Law: Disk density doubles every 13 months
- Best principal for small and simple applications -> Simple website with a few DB calls is not HW intensive
- Bandwidth usually grows slower than computer power (conservative telecom/reluctant users).
- Wirth's law (1995): Hardware gets faster but software complexity and resource demands grow even faster

1.3. Energy: The new scaling constraint

- AI-specific hardware will surpass bitcoin scaling-economics
- ChatGPT query: 0.3W

- 100 queries roughly 15min of netflix
- Efficiency is improving
- Time savings give indirect energy savings
- Rebound effect -> users generate more and more queries

2. Week 2 - Location, Latency and Redundancy

2.1. Distributed Systems Motivation - Location

- Hardware might get faster but latency stays
- Nothing is faster than the speed of light -> you will always have some latency
- Reduce latency by placing systems closer to users (or by using a CDN)
- Legal requirements for data location (GDPR)

Latency: time for signal to travel from source to destination and back (round-trip time)

Traceroute: Tool used to visualize hops

- Ping only shows total RTT
- Traceroute shows the path
- Sends package with increasing TTL (Each router makes TTL - 1 when TTL = 0 then ICMP "Time Exceeded" is sent)

2.1.1. Speed of Light

- There is a difference between practical and theoretical limit (300ms vs 110ms)
- Usually no direct paths from one location to another (e.g there is no straight cable from my home to my friends home in amsterdam)
- Signals can only travel at the speed of light inside a vacuum

2.1.1.1. Different fiber types

Single mode fibers: Provide lower latency, one main path, less delay spread

Multimode fibers: Many paths, some longer, more delay spread

The refractive index of glass depends slightly on the wavelength of the light. This is called chromatic dispersion. Different wavelengths travel at slightly different speeds, so the latency depends on what optical wavelength is used

Hollow core fiber: Has less latency, guides light through air and not through solid glass. Thus, light can travel closer to its vacuum speed.

Copper: Propagates faster but not much (Good copper cable > standard fiber)

Quick facts:

- Network latency has a physical lower limit because signals cannot travel faster than light.
- Fiber is slower than space/air because light travels through glass at about 200,000 km/s, not 300,000 km/s.
- Real network latency is higher than the theoretical minimum due to routing, queuing, protocol overhead, traffic inspection and signal repeating.
- Satellites can sometimes be faster than fiber because they can take a more direct route and signals travel faster through air/space. However, weather conditions can affect signal strength
- Wi-Fi is not the lowest-latency option because it adds waiting time, acknowledgements, retransmissions and MAC-layer processing.
- Main idea: latency depends on distance, signal speed, routing path and protocol overhead.

2.2. Motivation for Distributed Systems - Redundancy / Fault-tolerance

- Any hardware will crash eventually -> it's not a question of if but when
- Random bit flips in memory

  - Might happen for DNS names in your memory, that's why many providers have **Bitsquat Domains** (e.g ikamai.net -> akamai.net)
- Cosmic rays may be blamed for an electronic voting error (bit flip)
- Techniques like replication, checksums, redundancy, consensus and monitoring help keep systems reliable.
- HDD break, SSDs wear out (use NAND cells with limited lifetime, they have spare NAND)

-Distributed Systems provide a solution
- Multiple machines provide redundancy
- When one machine fails, others can take overhead
- Load can be redistributed among remaining machines
- System continues to function despite individual failures

Trade-off consideration:
- Distributed systems add complexity
- Use only when benefits outweigh the added complexity
- Redundancy and fault tolerance must justify the complexity

One possible solution:
- Error-correcting code memory (ECC)
- Hamming Code, correct 1 bitflip / detect 2 bitflips
- Used mainly for servers
- Consumer: DDR5 has on-die ECC but weaker than traditional ECC (corrects only inside the chip at rest but not on the data bus in transit)

NAND flash memory types:
- SLC = Single-Level Cell -> Stores 1 bit per cell.
- MLC = Multi-Level Cell -> Usually stores 2 bits per cell.
- TLC = Triple-Level Cell -> Stores 3 bits per cell.
- QLC = Quad-Level Cell -> Stores 4 bits per cell.

The tradeoff is: more bits per cell means cheaper and denser storage, but lower endurance and usually slower writes. Caching with SLC → files / cells that are frequently changed, store on SLC, once they don't change that often move to MLC/TLC/QLC

The goal is to level the wear and distribute write and erase operations across all memory cells.

2.3. Fault Tolerance

Seacables provide 99% of data connectivity. Network outages do happen from time to time.

3. Key takeaways (Week 1-2) - Why do we need Distributed Systems

3.1. Why do we need distributed systems?

We need distributed systems mainly for scaling, location, and fault-tolerance. Distributed systems add complexity, so this should be avoided unless needed.

3.1.1. Scaling (if one machine is not enough)

Scaling means increasing system capacity.

Vertical scaling means using a stronger machine, for example more memory or a faster CPU.

Advantages:
- Lower cost at small scale
- No software adaptation required
- Less complexity

Disadvantages:
- Hardware limits
- Hardware failure can cause an outage
- Fault-tolerance is harder
- Speedup is limited by sequential work, see Amdahl's Law

Horizontal scaling means using more machines.

Advantages:
- Lower cost at massive scale
- Easier to add fault-tolerance
- Higher availability

Disadvantages:
- Software adaptation required
- More complex system with more components

3.1.2. Location (to move closer to the user)

Location matters because hardware can get faster, but latency remains limited by the speed of light. Distributed systems can reduce latency by placing systems closer to users, for example with a CDN.

Other reason for location:
- Legal requirements for data location, such as GDPR

Latency is the time for a signal to travel from source to destination and back.

Important points:
- Real network paths are usually not direct
- Fiber is slower than vacuum because light travels through glass
- Single-mode fiber has lower latency than multimode fiber
- Hollow-core fiber can have lower latency because light travels mostly through air
- Copper can be slightly faster than standard fiber, but not by much
- Traceroute shows network hops, while ping only shows total RTT

3.1.3. Fault-tolerance (HW will fail eventually)

Fault-tolerance means preparing for failures. Hardware will eventually fail, and at scale rare failures become expected.

Examples:
- Random bit flips in memory
- HDDs break
- SSDs wear out
- Network outages happen, including sea cable failures

Distributed systems help because:
- Multiple machines provide redundancy
- If one machine fails, others can take over
- Load can be redistributed
- The system can continue despite individual failures

Reliability techniques from Week 2:
- Replication
- Checksums
- Redundancy
- Consensus
- Monitoring
- ECC memory

4. Week 3

4.1. Key takeaways

4.1.1. What is the difference of VM / Container?

A VM runs like a real computer with its own guest OS on a hypervisor. A container is an isolated user-space instance that shares the host OS/kernel.

4.1.2. How does docker work (container implementation)?

Docker packages software into containers. Containers are isolated with Linux mechanisms like network namespaces, cgroups, OverlayFS, seccomp profiles, and Linux capabilities.

4.1.2.1. Best practices

Keep images small and up to date, scan for vulnerabilities, do not expose the Docker daemon socket, do not run as root, limit capabilities/resources, and use read-only filesystems/volumes where possible.

4.1.3. What is docker-compose, and how to run multiple services

Docker Compose runs multiple containers together, for example services, clients, databases, or load balancers.

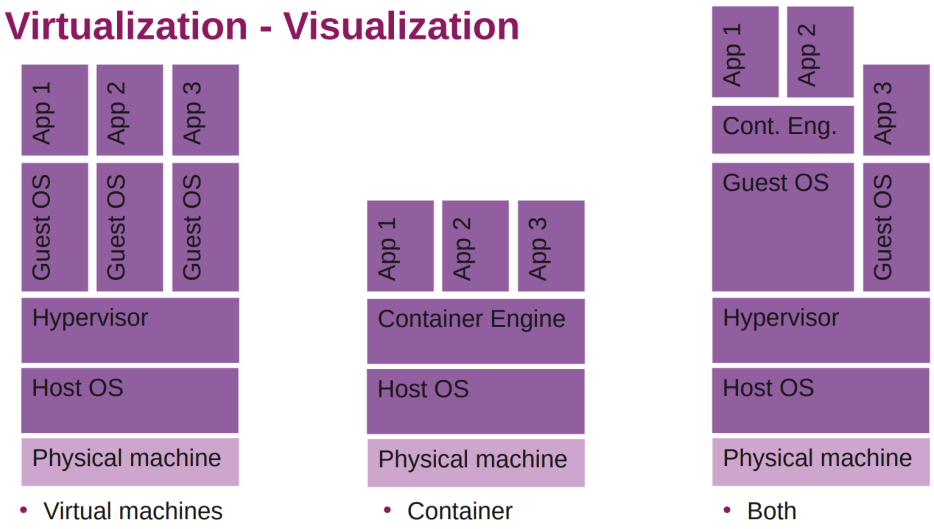
Define the services in a `docker-compose.yml`, then Docker Compose starts and connects them as one application setup.

4.2. Containers and VMs

4.2.1. Virtualization

Creation of a virtual machine that acts like a real computer with an operating system.

- **Host Machine:** machine where the virtualization software runs
- **Guest machine:** Virtual machine
- **Hypervisor:** Runs virtual machine
  - Type 1: bare metal e.g VMware ESXi
  - Type 2: hosted e.g VirtualBox
- Run a modified OS with Intel VT-x and AMD-V, or paravirtualized if not present (VM should not access memory directly)
- Needs to be the same architecture (otherwise needs to be emulated)
- Virtual Desktop Infrastructure (VDI)
- Containers
  - Isolated user-space instances
  - OS support: isolations



4.2.2. VM vs Container

+ Reduced size of snapshots 2MB vs 67MB
+ Quicker spinning up apps
+/- Available memory is shared
+/- Process-based isolation (share same kernel)

Use Case: complex application setup, with container less complex configuration Providers: ECS, Google Cloud Run, Digital Ocean App Platform

+ App can access all OS resources
+ Live migrations
+/- Pre allocates memory
+/- Full isolation

Use Case: better hardware utilization / resource sharing Providers: EC2, Virtual Machines, Computer Engine, Droplets

4.2.3. Prices

Cloud VM pricing has several models:

- On-demand: Pay as you go for the VM plus related costs like data transfer and IP addresses.
- Spot instances: Cheaper, discounted compute capacity when providers have unused resources, but availability is less reliable.
- Reserved instances: Lower prices in exchange for committing to usage over a longer period.

Main takeaway: VM cost optimization requires regularly comparing providers and pricing models, since costs and resource bundles differ and can change over time.

4.2.4. Firecracker vs VirtualBox

Firecracker is a lightweight MicroVM with minimal devices, no BIOS/UEFI boot, very fast startup and low memory overhead.

VirtualBox is a full type 2 VM with broad device emulation, a normal BIOS/UEFI boot chain, slower startup and pre-allocated memory.

Main takeaway: Firecracker is optimized for speed and efficiency, VirtualBox is optimized for full VM functionality.

4.3. Docker

4.3.1. Container Introduction

- **LXC (Linux Containers):** Lower abstraction level and direct use of Linux kernel features
- **systemd-nspawn:** Part of the systemd project, minimalist container manager

- **Solaris Zones:** Oracle/Sun-specific container technology
- **Linux-VServer:** Kernel patch for Linux, older virtualization technology
- **OpenVz:** Operating system-level virtualization, popular hosting tool
- **Singularity:** Scientifically oriented container solution, HPC-friendly
- **Podman:** Docker alternative / replacement
- **Docker:** / docker-compose for software development

4.3.2. Docker Introduction

- Docker is a containerization platform that packages software into containers (existing images can be found on docker hub)
- Containers are isolated from each other with linux-based isolation mechanisms, they only communicate over well-defined channels
  - Network namespaces
    - Each container gets its own network environment (e.g IP, port, interfaces)
  - Cgroups
    - Control how many resources a container can use (e.g CPU, RAM, disk I/O)
  - OverlayFS
    - Lets docker build images in layers
    - Container can share read-only image layers and add a small writable layer on top
  - Security
    - seccomp profiles
      - Restrict which linux system calls a container is allowed to use
    - Linux capabilities
      - Split root privileges into smaller permissions
- Docker can be used on macOS or Windows
  - WSL2 runs a lightweight linux VM, has better isolation than bare metal linux but I/O is slower

4.3.3. Docker Examples

- If no image version is specified, Docker uses the `latest` tag.
- Docker Hub is the main repository for container images.
- GUIs like Docker Desktop or Podman Desktop can also manage Docker.

- Install Docker and test it:
  - `docker run hello-world`
- List installed images:
  - `docker images`
  - `docker images -a`
- Remove an image:
  - `docker rmi hello-world`
  - `docker rmi <image-id>`
- Save and inspect an image:
  - `docker save hello-world -o test.tar`
  - `tar xf test.tar`
  - `tar xf <layer-id>/layer.tar`
- Run extracted program:

  - `./hello`
- List containers:
  - `docker ps -a`
- Remove a container:
  - `docker rm <container-id>`

4.3.4. Details

- Docker images use filesystem layers (FS Virtualization).
- OverlayFS combines multiple layers into one visible filesystem.
- Example Dockerfile:

```
FROM alpine
ADD hello.sh .
ENTRYPOINT ["sh", "hello.sh"]
```

- Example script (hello.sh):

```
#!/bin/sh
echo "Hallo"
```

- Meaning:
  - `FROM alpine`: use Alpine Linux as the base image.
  - `ADD hello.sh .`: copy `hello.sh` into the image.
  - `ENTRYPOINT ["sh", "hello.sh"]`: run the script when the container starts.
- Example commands:
  - `docker build . -t test`
  - `docker run test`
  - `docker save test:latest > test.tar`
- The image has at least two layers:
  - Alpine base layer, including BusyBox, musl libc, crypto/ssl, etc.
  - New layer containing `hello.sh`.
- If the input does not change, Docker reuses the cached layer.

4.3.5. OverlayFS

OverlayFS is a Linux filesystem that shows multiple directories as one combined directory.

It has three main parts:

- **lowerdir:** the base layer. Usually read-only. In Docker, this is like the image layer.
- **upperdir:** the writable layer. New or changed files are stored here.
- **workdir:** a temporary working directory OverlayFS uses internally when moving or updating files.
- **overlay mount:** the final merged view that the user/container sees.

Example:

```
cd /tmp
mkdir lower upper workdir overlay

sudo mount -t overlay -o \
lowerdir=/tmp/lower,\
upperdir=/tmp/upper,\
```

```
workdir=/tmp/workdir \
none /tmp/overlay
```

`/tmp/overlay` becomes the mount point and shows the combined contents of lower and upper

If a file exists in lowerdir and you edit it, OverlayFS does not modify the read-only lower layer. Instead, it copies the file into upperdir and changes it there.

If you delete a file that exists only in lowerdir, OverlayFS cannot really delete it from the read-only layer. So it creates a marker in upperdir saying “this file is deleted.” The file then disappears from the merged overlay view.

You can also have multiple lower layers:
`lowerdir=/tmp/lower1:/tmp/lower2 /tmp/overlay`

Main idea: OverlayFS lets Docker keep image layers read-only while storing container changes separately in a writable upper layer.

4.3.6. Cgroups

Control Groups: limits, isolates, prioritization of CPU, memory, disk I/O, network

Docker uses cgroups internally for container resource control.

- Example: create two CPU groups:

```
cgcreate -g cpu:red
cgcreate -g cpu:blue
```

- Assign different CPU weights:

```
echo -n "20" > /sys/fs/cgroup/blue/cpu.weight
echo -n "80" > /sys/fs/cgroup/red/cpu.weight
```

- Run shells inside the groups:

```
cgexec -g cpu:blue bash
cgexec -g cpu:red bash
```

- Test CPU competition on one core:

```
taskset -c 0 sha256sum /dev/urandom
```

- Docker example:

```
docker run --name=low_prio --cpuset-cpus=0 --cpu-shares=20 alpine sha256sum /dev/
urandom
docker run --name=high_prio --cpuset-cpus=0 --cpu-shares=80 alpine sha256sum /dev/
urandom
```

Main idea: cgroups let Linux and Docker isolate, limit, and prioritize resources between processes or containers.

4.3.7. Separate networks

Linux network namespaces provide isolation of the system resources associated with networking.

Create a namespace:

```
ip netns add testnet
ip netns list
```

Create a virtual Ethernet pair between host and namespace:

```
ip link add veth0 type veth peer name veth1 netns testnet
ip link list
```

Run commands inside the namespace:

```
ip netns exec testnet <cmd>
```

Configure IP addresses and enable interfaces:

```
ip addr add 10.1.1.1/24 dev veth0
ip netns exec testnet ip addr add 10.1.1.2/24 dev veth1
ip link set dev veth0 up
ip netns exec testnet ip link set dev veth1 up
```

Run a server inside the namespace:

```
ip netns exec testnet nc -l -p 8000
```

Connect the namespace to the outside network:

```
echo 1 > /proc/sys/net/ipv4/ip_forward
ip netns exec testnet ip route add default via 10.1.1.1 dev veth1
iptables -t nat -A POSTROUTING -s 10.1.1.0/24 -o enp6s0 -j MASQUERADE
iptables -A FORWARD -j ACCEPT
```

Main idea: network namespaces give containers separate network environments while still allowing controlled communication with the host or internet.

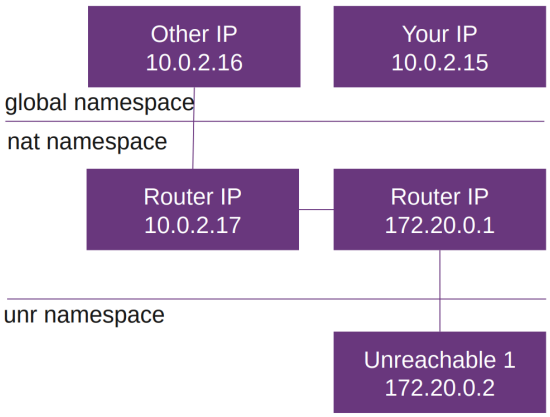
4.3.8. Connectivity, Security and Robustness

- **NAT (Network Address Translation):** Multiple devices share one public IP -> devices are not directly reachable
- **Hole punching:** Two peers behind NAT send coordinated packets to create NAT mappings, enabling a direct connection
  - P2P / Hole Punching development (in the old days)
  - Currently: network namespaces

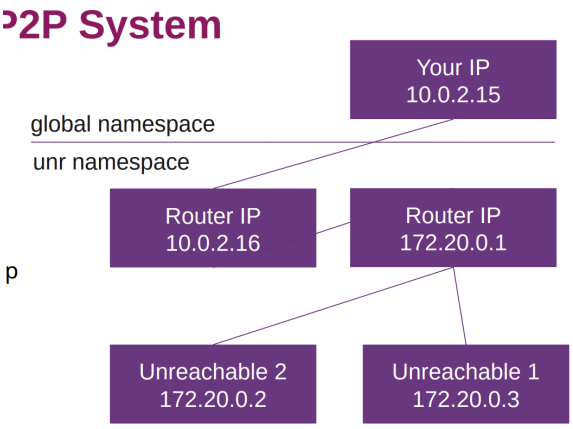
4.3.8.1. P2P System

- **veth**: Virtual Ethernet Device, like a virtual network cable between namespaces.
- Used to build a local P2P testbed with peers that are reachable or hidden behind NAT.
- Example setup: global namespace `10.0.2.15`, router/NAT `10.0.2.16` and `172.20.0.1`, unreachable peers `172.20.0.2` and `172.20.0.3`.
- Create namespace, veth pair, and router interface:
  - `unr` is the isolated network namespace for the unreachable peer side.
  - `nat_lan` and `nat_wan` are two ends of one virtual Ethernet cable.
  - `nat_lan` is moved into `unr`, while `nat_wan` stays in the global namespace.

```
ip netns add unr
ip netns list

ip link add nat_lan type veth peer name
nat_wan
ip link set nat_lan netns unr

ip address add 10.0.2.16/24 dev nat_wan
ip link set nat_wan up

ifconfig
ping <ip>

ip netns exec unr ip address add
172.20.0.1/24 dev nat_lan
ip netns exec unr ip link set nat_lan up
```



- Add unreachable peer interface and routes:
- `unr1` is a dummy interface that represents an unreachable peer inside the namespace.
- `172.20.0.2` is the peer IP, while `172.20.0.1` is the router/gateway inside `unr`.
- The loopback interface `lo` is enabled because many programs expect localhost to exist.
- The route commands tell traffic how to leave the namespace and how the host can reach the namespace network.




```
ip netns exec unr ip link add unr1 type
dummy
ip netns exec unr ip address add
172.20.0.2/24 dev unr1
ip netns exec unr ip link set unr1 up

ip netns exec unr ifconfig

ip netns exec unr ip link set lo up
ip netns exec unr route add default gw
172.20.0.1

ip route add 172.20.0.1 dev nat_wan
```

Main idea: network namespaces and veth pairs can simulate real P2P/NAT situations on one machine.

4.3.9. Docker Compose

Dockerfiles build individual container images. Docker Compose runs multiple containers together as one application setup.

- Dockerfile:
  - Used to build your own image.
  - Keep images small by only copying required files.
  - Docker caches layers aggressively, so unchanged steps are reused.

  - Squash images: `COPY --from=initial / /`

- Example multi-stage Dockerfile:

```
FROM golang:alpine AS builder
WORKDIR /build
COPY server.go .
RUN go build server.go

FROM alpine
WORKDIR /app
COPY --from=builder /build/server .
ENTRYPOINT ["./server"]
```

- Meaning:
  - First stage `builder`: contains Go and builds the binary.
  - Second stage `alpine`: small final runtime image.
  - `COPY --from=builder`: copies only the compiled binary into the final image.
- Docker Compose:
  - Used to deploy multiple containers, e.g. server, client, database, load balancer.
  - Configures services in one `docker-compose.yml`.
  - Provides lightweight orchestration, e.g. starting services that depend on others.
- Example `docker-compose.yml`:

```
services:
  server1:
    build: .
  client:
    image: alpine
    entrypoint: >
      sh -c "sleep 3 && echo hallo | nc server1 8081"
```

- Meaning:
  - `server1` is built from the local Dockerfile.
  - `client` uses the Alpine image.
  - The client waits briefly, then sends `hallo` to `server1` on port `8081`.

Main idea: Dockerfiles define images. Docker Compose defines how multiple containers are started and connected.

4.3.10. Docker Security

Best Practices:

- Keep images small, up to date / attack surface
  - **alpine / distroless:** Alpine is preferred because it creates very small, minimal Docker images that are faster to transfer, have fewer unnecessary packages and usually a smaller security surface.
- Check your image for vulnerabilities, snyk, trivy
- Do not expose the Docker daemon socket (even to containers)
- Set a user - do not run as root
  - Needs more configuration but good advice
- Limit capabilities (grant only what is needed by a container) -> seems overkill
- Limit resources (memory, CPU, file descriptors, processes, restarts)

  - Good for production -> docker does not have a hard memory limit. Docker kills process that goes over limit, no pressure for GC if not docker aware
- Set file system and volumes to read-only

4.3.11. How to debug

What is your base image:

1. Ubuntu, Debian, Alpine, distroless etc.
2. GNU libc (e.g distroless) vs. musl (e.g alpine)
   - Glibc (GNU C library): wide adaptation, many distros use it, larger binary
   - Musl (small C standard library): less used, smaller binary
   - If glibc is needed in alpine then a compatible busybox docker image or installing gcompat can help
   - Building outside docker container might fail because of missing libraries etc.

Alpine Tools:

- Busybox: nc, ping, sha256sum, wget, netstat
- Reach other container: ping containername
- Service bound to localhost? Cannot run outside docker, use 0.0.0.0

Other stuff:

- docker system prune -a (attention)
  - Deletes unused docker data to free disk space (stopped containers, unused networks, unused images, build cache)
- Check logfiles
- Performance: multi-stage, .dockerignore
- Docker: aggressive caching - wget is cached, use version numbers

5. Week 4

5.1. Key takeaway

5.1.1. What is a monorepo, what is a polyrepo?

Monorepo: One repository contains multiple parts of a project, e.g. frontend and backend in separate folders.

Polyrepo: A project is split across multiple repositories, e.g. one repo for frontend and one for backend.

5.1.2. When to use which type?

When to use monorepo: Good when projects are tightly connected, need shared code, or often change together.

When to use polyrepo: Good when projects are independent, need separate access control, or are managed by different teams.

5.1.3. How do AI coding agents change the monorepo vs. polyrepo decision?

Monorepos are easier for agents because they can see all related code at once. Polyrepos need extra setup and more context management.

5.2. Project setup

General Rules:

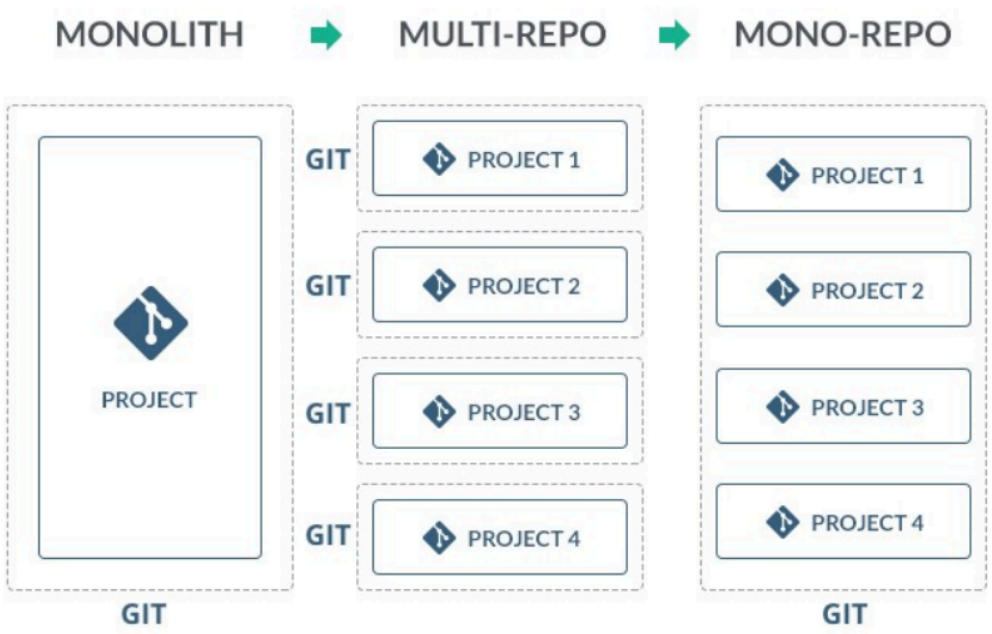
1. Be consistent
2. Other projects always prefer other structures
3. A perfect structure does not exist
4. Every project has a readme
5. Follow best practices (e.g java in src/main/java)

Split up:

- Backend, frontend in separate repositories
- 1 technology per split for simple projects
  - Most likely you won't have a frontend mix of frontend technologies e.g, Angular with Vue
  - Sometimes you have a script directory with different languages (bash, javascript)
- Keep config out of code (secrets, API keys -> .env not committed)
- More complex setups -> multiple backends, multiple packages

5.3. Monorepo (OneRepo / UniRepo)

- One repository for all projects
  - 1 sub-directory with frontend, 1 sub-directory with backend etc.
- Easier setup for AI agents



5.4. Polyrepo (Manyrepo / Multirepo)

- Multiple repository for a project
- Frontend in different repo than backend
- Sync via git submodules or via bash script
  - Submodules: can also be used as dependency management
- Sync with repo or via bash script

5.5. Pro/Cons

Monorepo:

- Tight coupling of projects
  - E.g., generating openapi.yml from backend, generate types for frontend → simply copy, or tRPC
- Everyone sees all code / commits
- Encourages code sharing within organization
- AI agents: full cross-project context (backend + frontend in one PR)
- Atomic commits for cross-project changes
- Scaling: large repos, specialized tooling
- Risk: cross-package dependencies (one change can affect many service -)

Polyrepo:

- Loose coupling of projects
  - If you want to generate openapi.yml, you need access from the backend repository to the frontend (e.g., curl+token)
- Fine grained access control
- Encourages code sharing across organizations
- AI agents: possible (–add-dir), but extra setup and token cost
- Synthetic Monorepo (Nx): bridges repo boundaries without moving code
- Scaling: many projects, special coordination

5.6. Tools

- Monorepos need tools to manage builds, tests, and dependencies across many packages.
- Tools include pnpm workspaces, Turbo, Nx, Gradle, Bazel, Buck2, or simpler scripts like make/just.
- A key feature is caching: if inputs did not change, the command does not need to run again.
- Main goal: speed up builds and reduce repeated work.

6. Week 5 - Load Balancing

6.1. Key takeaways

6.1.1. What types of LB exists?

Hardware, software, and cloud-based load balancers. Software load balancing can happen at L2/L3, L4, L7, DNS level, or with Anycast.

6.1.2. Which one to pick?

It depends on control and use case: hardware load balancers are mainly useful if you control your datacenter; cloud load balancers are pay-per-use with many offerings; software load balancers give options like Traefik, Caddy, Nginx, HAProxy, MetalLB, and DNS/Anycast approaches.

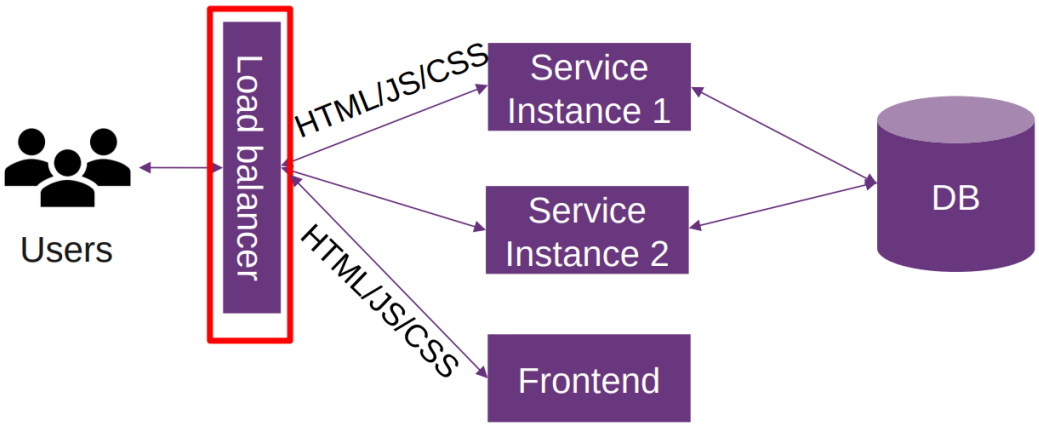
6.1.3. What is currently “state-of-the-art”?

Modern software load balancers/reverse proxies such as Traefik, Caddy, Nginx, and HAProxy, plus cloud-based L4/L7 load balancers. Additionally, you should measure performance yourself with realistic, reproducible benchmarks.

6.1.4. CORS

CORS means Cross-Origin Resource Sharing. Browsers restrict cross-origin HTTP requests from scripts. A common solution is using a reverse proxy so frontend assets and API appear under the same origin. Another workaround is setting Access-Control-Origin during development.

6.2. Load Balancing



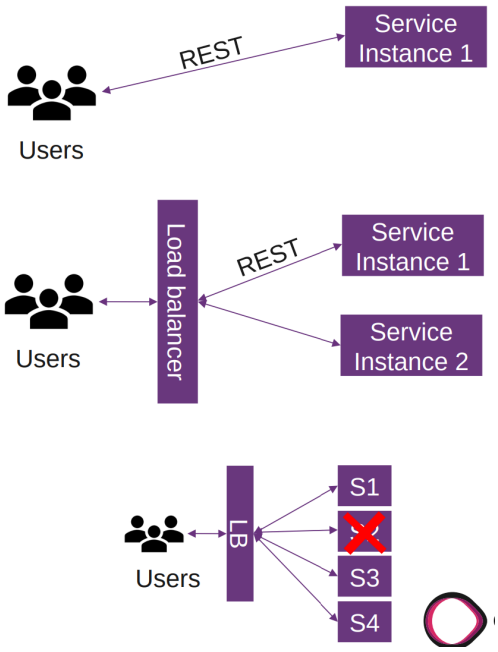
What is load balancing?

- Distribution of workload across multiple computing resources
  - Workloads (requests)
  - Computing resources (machines)
- Distributes client requests or network load efficiently across multiple servers
  - E.g service gets popular, high load on service

-> Horizontal scaling

Why load balancing?

- Ensures high availability and reliability by sending requests only to servers that are online
- Provides the flexibility to add or subtract servers as demand dictates



6.2.1. Types of load balancer (Cloud, Hardware, Software)

6.2.1.1. Hardware load balancer

- HW-LB use proprietary software, which often uses specialized processors
  - Less generic, more Performance
  - Some use open-source SW, e.g HA-proxy
- E.g loadbalancer.org, cisco, kemp
- Only if you control your datacenter

6.2.1.2. Software load balancer

- L2/L3: MetalLB (Kubernetes bare-metal, standard solution for type: LoadBalancer on bare-metal/ kind clusters without a cloud LB), Seesaw (not widely used)
- L4: HAProxy (desc), Nginx, Gobetween, Traefik
- L7: Enovy (C++), HAProxy (C), Nginx (C), caddy (golang), Gobetween (golang), Eureka (java) - services register at eureka, traefik (golang)

Measure performance yourself with realistic, reproducible benchmarks instead of relying only on opinions or generic benchmark results.

L4 vs L7 Load Balancing:

- L7: more resource intensive, can make smarter decisions
- L7: terminates TLS and inspects HTTP (encryption overhead)

DNS Load Balancing (L7):

- Round-robin DNS
  - Easy to setup, client-side selection
  - Drawback: negative caching impact
  - Used in Bitcoin Core: dig dnsseed.emzy.de
  - Example:

```
www A 188.40.119.115
lb  A 188.40.119.115
lb  A 152.96.80.48
```

- Split horizon DNS
  - Different DNS information, depending on source of DNS request
- Layer 3: Anycast
  - Multiple servers announce the same IP address from different location
  - Requires ownership of an AS and BGP routing

Main idea: software load balancing can happen at HTTP level, DNS level, or IP routing level, each with different complexity, performance, and control.

6.2.1.3. Cloud-based load balancer

- Pay for use, many offerings
- AWS
  - Application Load Balancer ALB (L7)
  - Network Load Balancer (L4)
  - Classic Load Balancer (legacy)
- Google Cloud (L3, L4, L7)
- Cloudflare (L4, L7)
- DigitalOcean (L4)
- Azure (L4, L7)

6.2.2. Load Balancing Algorithms

- Easiest: round-robin / random
  - distributing requests evenly by rotating through a list of servers.
  - Make sure your services are stateless
- Stateless: don't store anything in the service
  - If you do, you need a stick session (avoid this) - same user to same service
  - E.g cookie, ip_hash - send to same machine
- Health checks: tell your load balancer if you are running low on resources
  - Active: send active probes e.g every 3s
  - OOB - out of band (API to check health) e.g necessary with DB, as connection may be OK, but database not
  - Passive: only check with request
- Different behaviour:
  - Nginx: passive, caches requests, so if an upstream fails, it uses another
  - Caddy: passive, does not cache, but marks upstream as failed for the next request

Round-robin:

- Loop sequentially
- Simple algorithm, often default
- May drop requests on congested nodes

Weighted round-robin:

- You can put weighted in from everything
- Some servers are more powerful -> more powerful machines get more work
- But high variance in server load may drop requests

Least connections / fewest current connections to clients:
- Load balancer sends the request to the server that currently has the fewest outstanding requests
- Keep track of outstanding requests
- Not best for latency -> does not know if requests in queue are fast or slow

Peak exponentially weighted moving average:
- Measures average on how long a server takes to respond and gives importance to recent measurements
- Considers latency
- Increased complexity

6.2.3. Traefik
- Open Source, software-based load balancer
  - L4/L7 load balancer
  - Traefik is written in Go
  - Can support authentication mechanisms, especially for the dashboard
  - HTTP/3 support
- Has a built-in dashboard
- Traefik provides an official docker image

6.2.4. Service
- As a start -> is stateful
- Written in Go
- Stickiness with cookies
- Weighted round round-robin
  - Load balances between services and not between servers

6.2.5. Caddy
- Configuration: dynamic
  - Static: Caddyfile
- One liners:
  - Quick local file server: caddy file-server
  - Reverse proxy: caddy reverse-proxy –from example.com –to localhost:9000
- Open source, software-based load balancer
- L7 load balancer
- Reverse proxy
- Static file server
- HTTP/1.1, HTTP/2, HTTP/3
- Caddy is on docker hub
- Automatic HTTPS (Let’s Encrypt)

6.2.6. Nginx
- Free and commercial version
- Fast webserver, 35% market share
- HTTP proxy, mail proxy, reverse proxy, load balancer
- No active health checks, no sticky sessions

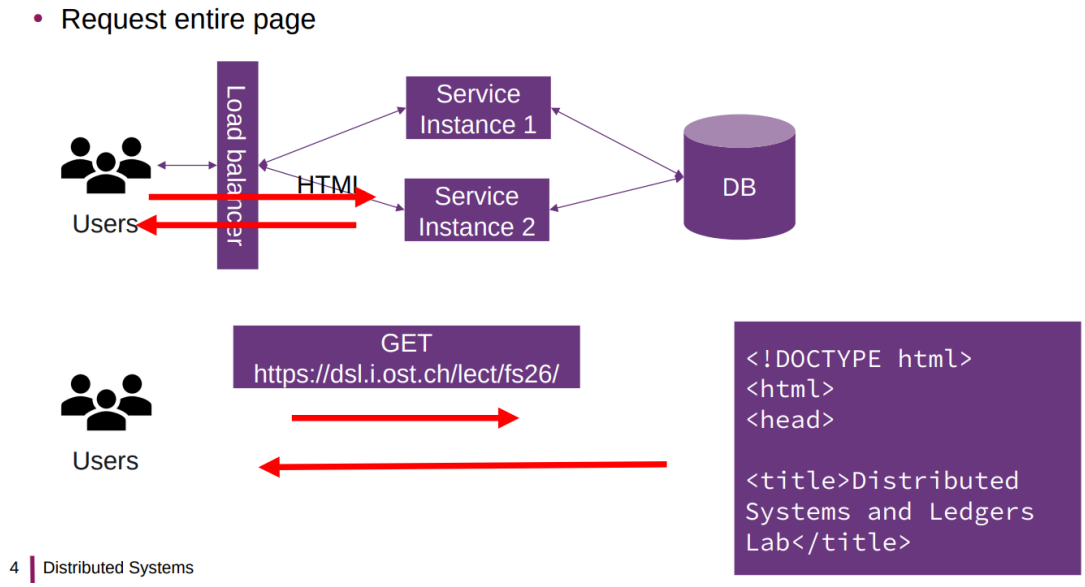
6.2.7. HAProxy
- L4 and L7 load balancer and reverse proxy
  - Open source option: commercial support

- Fast performance
- Configure and run: /etc/init.d/haproxy start
  - Algorithms: roundrobin, leastconn, source
  - Sticky session: cookie
  - check -> health checks (passive / active)
- Primary/secondary: backup server only receives traffic when all primary servers fail
- Dynamic backend discovery via server-template + DNS resolvers
- Built-in dashboard
- Rate limiting and connection throttling built-in

6.3. Web Architecture

6.3.1. Server-Side rendering

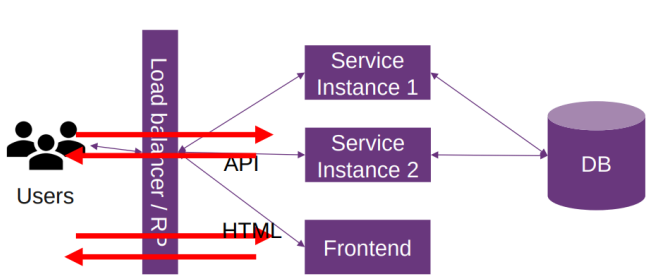
- Classic approach: SSR
- Server generates HTML/JS/CSS (dynamically)
  - User request: browser sends a request to the web server (server-side routing)
  - Server processing: server processes request by running server-side code (C#, Java, …)
    - May require data from database or other sources
    - Server-side code can use template engines or a framework to render the HTML
  - Response: Generate the appropriate HTML, CSS and JavaScript for the requested page
  - Browser rendering: browser receives response and renders page
- Advantage: SEO, immediate display
- Disadvantage: server rendering for every request (caching), UI logic on the server
- Static site generation (SSG): pre-render HTML/CSS/JS: only once, regenerate if content changes



6.4. Single Page Application SPA / CSR

- Interactions occur within a single web page
- App-like experience: client page dynamically updates as user interacts
- Relies on JavaScript to update UI, typically:
  - Initial response: server returns a single HTML file with references to CSS/JavaScript
  - Browser rendering: shows initial empty HTML file with a spinner, then executes JavaScript, then shows UI

- User interactions: JavaScript manages the UI updates. Application does not require full page reloads. When you click a link in an SPA, instead of making a traditional HTTP request:
  - JavaScript intercepts the click events
  - Prevent default browser navigation
  - Update the URL using the History API
  - Render new content without requesting new HTML document, but may involve fetching data
- Fetching data: When the SPA needs to fetch or send data, communicates through APIs
- Use a framework: React, Angular, Vue
- Backend serves API requests only
- SEO only works if JavaScript is executed
  - Crawler gets JavaScript code, needs to execute, then it knows the content
- Good separation: UI in HTML/CSS/JS, backend in /api
- Client-side routing: SPAs for navigation
  - Server side routing -> default to index.html
- Typical setup
  - `/ -> index.html`
  - `/user -> user.html`

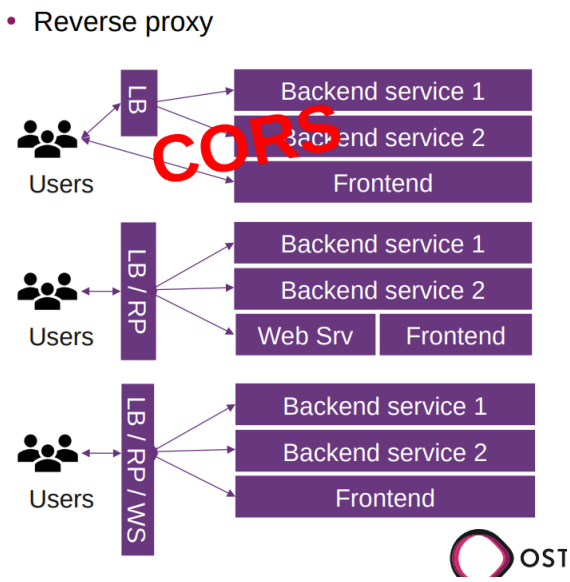


- Initial load: entire page
- Further requests: only updates partially


```
{"menu": {
  "id": "file",
  "value": "File",
...
```

6.4.1. CORS

- CORS = Cross-Origin Resource Sharing
- For security reasons, browsers restrict cross-origin HTTP requests initiated from scripts
- Mechanism to instruct browsers that run a resource from origin A to run resources from origin B
- Solution: reverse proxy with webserver
  - Many solutions, caddy, vite/webpack dev server, nginx, mix -> the client only sees the same origin for the API and the frontend assets
- "Workaround": Access-Control-Origin: https://…
  - For dev: Access-Control-Origin: *
  - Pre-flight requests
  - Golang: w.Header().Set("Access-Control-Origin", "*")



6.4.2. SSR vs CSR

Performance Metrics:

- TTFB (Time to First Byte): Server response time
- FCP (First Contentful Paint): When content first appears
- TTI (Time to Interactive): When page becomes fully interactive
- Trade-off: SSR good at FCP/TTI, CSR requires full JS execution first

SSR vs CSR Initial Load:

- SSR: visible and interactive immediately (no JS needed)
- CSR: Must download, parse, execute JS before interactive
- Post-load: CSR good: no page reloads for navigation, feels like desktop appears

CSR Advantages:

- Lower server rendering load
- API only serves JSON

CSR Disadvantage:

- Bundle Size Problem: Large JS files, slow parse/execution, mobile may struggle
- Slow initial load: white screen until JS executes
- SEO problem: crawlers see empty html, need JS executing to read content

Why CSR? Clean architecture (frontend/backend separation)

6.4.3. CSR Improvements

- Code splitting, lazy loading, hybrid approach with hydration (when a page is first sent as HTML but still has to load JavaScript to make HTML interactive)
  - Hydration problems:
    - Duplicated work / complexity
    - Pre-rendering only: PrevelteKit
- Server Components (React)
  - Components render only on server, no JS sent to client

  - Reduces bundle size
- Server HTML fragments htmx: server replies with HTML fragments
- Island architecture
  - Static HTML with interactive islands
  - Only islands ship JS - minimal JS by default (Astro)
- Streaming SSR
  - Send HTML in chunks as ready
  - Browser renders earlier - improves perceived performance
- Edge rendering
  - Render closer to user geographically at CDN edge locations (Cloudflare Workers, Vercel Functions)

7. Week 6

7.1. Key takeaways

7.1.1. Explain what MCP is, why it exists, and how tool calling works (tools/list, tools/call, JSON-RPC)

MCP is an open protocol by Anthropic that connects AI agents to external tools, data sources, and services. It exists to avoid every AI app needing custom connectors for every tool. Tool calling uses JSON-RPC 2.0:

- tools/list: asks what tools are available
- tools/call: runs a selected tool with arguments

7.1.2. Describe the MCP message flow between user, client, LLM, and MCP server

The flow is: `user -> assistant (tool_call) -> tool (result) -> assistant (answer)`.

The LLM decides whether to call a tool, sends the tool call, receives the result, and then creates the final answer. Tool results must include the correct `tool_call_id`.

7.1.3. Discuss challenges of running MCP in the browser and how WebMCP addresses them

Browsers can block or limit external fetching, causing errors like `"could not fetch"`.

WebMCP solves this by using browser-side tools, for example a Firefox extension that injects a `client_web_search` tool and acts like an MCP server.

7.1.4. What authentication mechanisms exists for web applications?

- Passwords / single-factor authentication
- MFA / 2FA
- TOTP software tokens
- Basic Auth
- Digest Auth
- mTLS
- Session-based authentication
- JWT
- Access + refresh tokens
- OAuth

7.1.5. How can stateless authentication be achieved?

Stateless authentication can be achieved with JWTs.

After login, the server creates and signs a token. The client sends it on future requests with `Authorization: Bearer <token>`. Because every server instance can verify the token, the server does not need to remember a session.

7.2. Model Context Protocol (MCP)

- By anthropic, november 2024, open-source
  - Claude Opus / Sonnet
- Donated to linux foundation -> no single company controls it
- Connects AI agents to external tools, data sources and services
  - Universal Protocol
  - Implemented once, works everywhere
- Before MCP: every AI app needed a custom connector per tool (NxM problem)
  - Doesn’t scale, fragile, duplicated effort

- Born from a single devs frustration copy-pasting between claude desktop and his IDE
  - Not a corporate strategy, solving an annoying workflow problem

7.2.1. MCP API

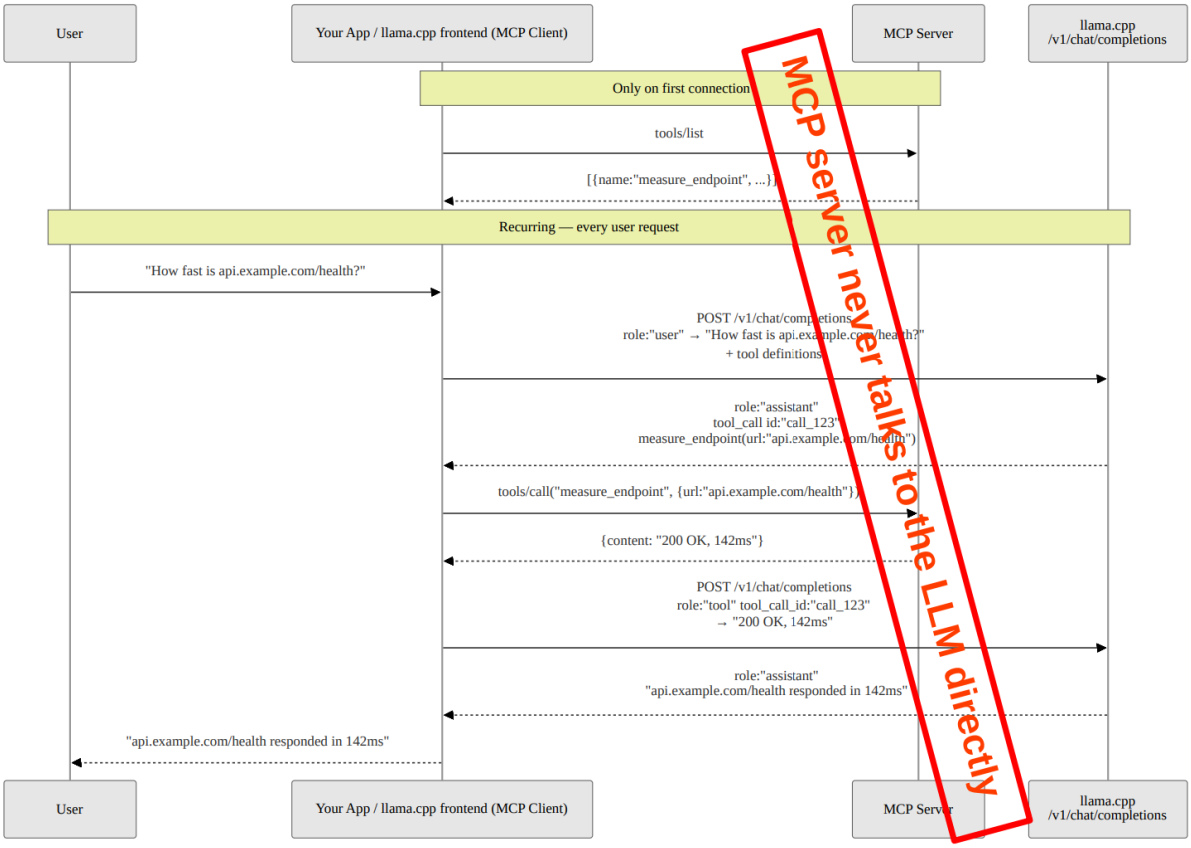
The MCP API uses JSON-RPC 2.0 for communication between an MCP client and an MCP server.

First, the client sends a `tools/list` request. This asks the server which tools are available. The server responds with an array of tools. Each tool contains a name, a description and an input schema that defines which arguments are required.

Afterwards, the client can use `tools/call` to execute one of these tools. The request contains the tool name and the required arguments. The MCP server then runs the tool and returns the result.

In short, `tools/list` is used to ask “what can you do?”, while `tools/call` is used to say “run this tool with these arguments”.

7.2.2. MCP Sequence Diagram



7.2.3. Simple MCP Server

A simple MCP server can expose tools that an LLM is allowed to call. In this example, the server provides one tool called `measure_endpoint`, which measures the response time of a given URL.

The available tools must be included in the LLM request. Each tool has a name, a description and a parameter schema. Based on the user request, the LLM decides whether a tool should be called, which tool to call and which arguments to use.

The usual role flow is:

```
user -> assistant (tool_call) -> tool (result) -> assistant (answer)
```

This means the user first asks a question, the assistant decides to call a tool, the tool returns the result and the assistant uses that result to generate the final answer.

The OpenAI Chat API is stateless, which means previous tool calls and their results are not remembered automatically. Therefore, after the tool has been executed, a second call to the LLM is required. This second request includes the original user message, the assistant message containing the tool calls and the tool result messages.

Each tool result must reference the correct `tool_call_id`. The LLM generates this ID when it requests a tool call and the application must echo it back in the matching tool response. If multiple tools are called, multiple `role: "tool"` messages must be returned, each with its own matching `tool_call_id`.

Using the wrong role or an incorrect `tool_call_id` can lead to errors or incorrect generated results, because the LLM can no longer reliably match tool outputs to the requested tool calls.

7.2.4. MCP in the Browser

- The client sends the available tools together with the user query.
- The LLM returns tool calls with names and arguments.
- The tool result is sent back to the LLM to generate the final answer.

Problem: A normal LLM running in the browser cannot reliably access external websites or fetch linked resources by itself. Browser restrictions, missing tool access or rate limits can cause errors such as “could not fetch”. Therefore, the model needs a way to use browser-side tools.

- A Firefox extension can intercept API requests and inject a `client_web_search` tool.
- The extension acts like the MCP server, so no separate MCP server is needed.
- This allows the LLM to search or fetch web content through the browser.
- In the future, WebMCP could allow websites to expose tools directly through the browser.

7.3. Authentication

- Single-factor authentication, e.g password
- Multi-factor authentication / 2FA, e.g password and software token, SMS unsecure
- Password rules - don’t:
  - Name of a pet, child, family member, or significant other
  - Anniversary dates and birthdays, birthplace
  - Name of a favorite holiday
  - The word “password”
- Don’t reuse passwords, use password managers
- Don’t enter passwords on unencrypted sites
- Use Argon2id for password storage
- Hashtype: WPA/WPA2
- Software token: TOTP (Time-based One-time Password)
  - Often used as 2nd factor (e.g Google Authenticator)
  - Based on keyed-hash message authentication code
  - hash(key + message)
  - K = shared secret
  - T = current unix time / 30s
  - TOTP(K, T) = Truncate(HMAC-SHA-256(K,T))
- Where should auth happen?
  - In service, e.g your HTTP server
  - In load balance, e.g traefik, jwt

7.3.1. Information security - key concepts / access control

- Confidentiality
  - Protects against eavesdroppers
- Integrity
  - Protection against data modification
- Availability
  - Data needs to be available when needed
- Non-repudiation
  - Neither the sender nor the receiver can deny that a communication has taken place
- Identification
  - E.g with a username “alice”, claiming to be Alice
- Authentication
  - Verifying a claim of identity, e.g alice shows passport, authentication types:
    - Something you know: things such as a PIN, a password
    - Something you have: a key, a swipe card
    - Something you are: biometrics: fingerprint
- Authorization
  - What resources an authenticated user is permitted to access

7.3.2. Basic Auth

Client sends a username and password with a HTTP request.

- Should be used with HTTPS
  - With basic auth the password isn’t encrypted
  - Password gets Base64 encoded
    - Base64: encoding format, not security. Can be decoded by anyone
  - Client will provide something like: `Authorization: Basic <base64>`
- Can be encoded in URL
- Server will reply with header
  - WWW-Authenticate: Basic realm=“restricted area”
  - The user will see the information “restricted area”

7.3.3. Digest Auth

- Hash + nonce, against replay attacks
- Server send
  - `WWW-Authenticate: Digest realm="testrealm@host.com"`
- Client sends in HTTP header
  - `Authorization: Digest username="Alice"realm="testrealm@host.com", nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093", uri="/dir/index.html"`
- Advantages
  - PW not in clear text (MD5), can be “SHA-256”, “SHA-256-sess”, “SHA-512” and “SHA-512-sess”
    - sess: “session key” for “authentication session”
  - Nonce for replay protection for client and server
- Disadvantages
  - Browser L&F
  - Cannot use scrypt or bcrypt to store PWs

7.3.4. mTLS

Basic explanation: With mTLS both sides verify each other. e.g:
Client checks: Is this really the server?
Server checks: Is this client allowed?

- Create SSL CA certificates for server with openssl command
- Create CA / server cert
- Create client certificate
- Add caddy security in your local network (tls_client_auth)
  - If the certificate is missing or invalid inside caddy then the request is rejected

7.3.5. Lets Encrypt

- Free, automated, open certificate authority
- Provides domain control via challenges
- HTTP-Challenge verification
  - Server must be publicly reachable
  - Token replaced at /.well-known/acme-challenge/
  - Let's encrypt verifies token via HTTP request
- DNS-Challenge verification
  - Server does not need to be publicly reachable
  - Enables wildcard certificates (valid for all subdomains)
  - Token placed as TXT record in DNS
  - Let's encrypt verifies the DNS entry
- Certificates valid for 90 days (currently)
  - May 2026: opt-in 45-day certificates
  - Feb 2027: default 64-day certificates
  - Feb 2028: default 45-day certificates
  - Optional: 6-day short-lived certificates (since 2026)
- Integrated in modern web servers (e.g caddy)

7.3.6. Session-based authentication (stateful)

- The server remembers that a user is logged in.
- This can lead to problems with multiple service instances because the session is only stored in the memory of that instance
- E.g SpringBot (JSESSIONID)

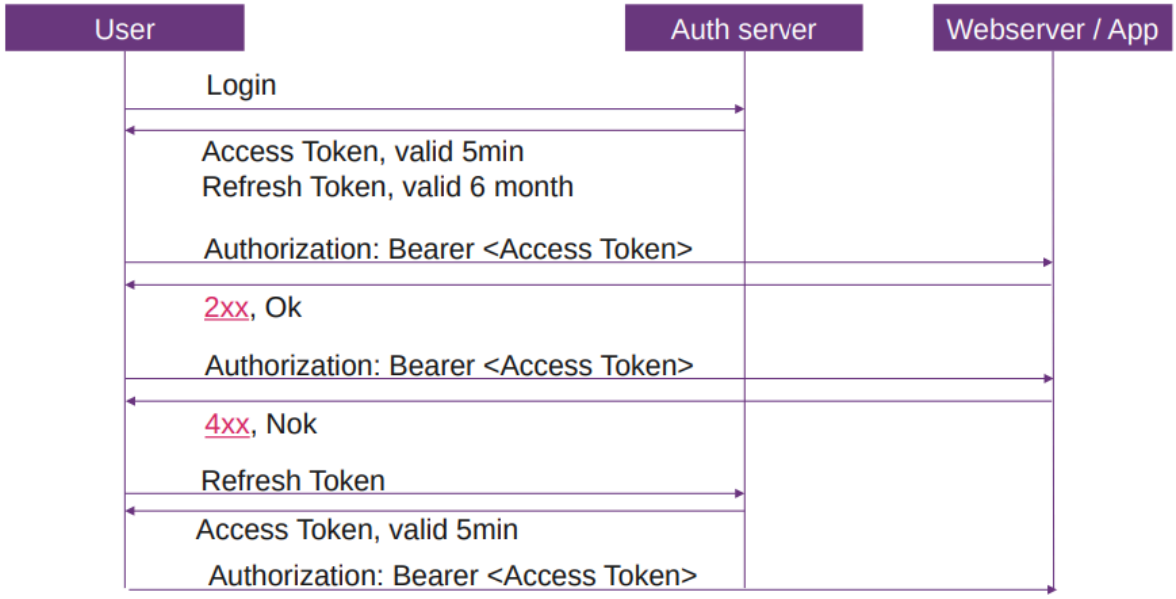
7.3.7. JSON Web Token (JWT) (stateless)

- JSON-based access tokens
  - Header: {"alg":"HS256"}
  - Payload: {"sub":"tom", "role":"admin","exp":"1422779638"}
- All server instances know a secret token / public key
- When user logs in, server send back token
- Client sends: Authorization: Bearer
- Token: const_user_token = base64urlEncoding(header) + '.' + base64urlEncoding(payload) + '.' + base64urlEncoding(signature)
- Signature (simple): keyed-hash message hash(base64(header)+base64(payload) + secret token)
- Client can store token in `localStorage.setItem("token", accessToken);`

Flow:

1. User logs in with username/password
2. Server checks if login is correct
3. Server creates a JWT
4. Server signs the JWT with the secret key
5. Server sends the finished JWT back to the client
6. Client sends this JWT with future requests

7.3.8. Access Token / Refresh Token



Access Token:
- Short lifetime, e.g 10min
- If public key / secret is known, the content in the token can be trusted
- Can have userId, role etc -> no need to query DB for those Information

Refresh Token:
- Longer lifetime, e.g 6 months
- Used to get a new access token
- IAM / Auth server creates access tokens

Only access token: If a user credential is revoked, how is every service informed?

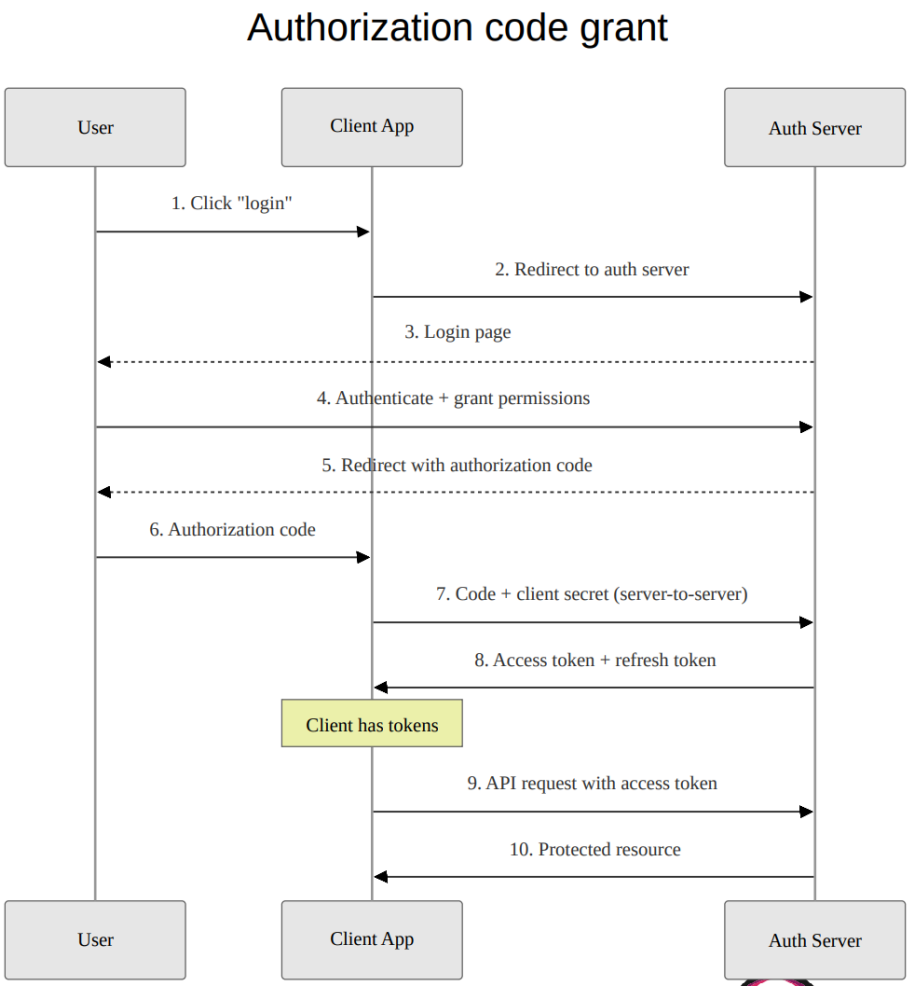
Only refresh token: For every request to Service, Auth needs to be involved for access

Access + Refresh token: If a user credential is revoked, user has max. 10 min more to access service, but auth is only involved if the access token is expired

7.3.9. OAuth

- For authorization with 3rd party integration
  - Grant access without giving away passwords
- Authorization code grant
  - User redirected to authorization server
  - User authenticates and grants permission
  - Authorization code returned to app
- PKCE: variant for clients that cannot store a client secret securely (mobile, SPA)

- Token issued are typically JWTs (access + refresh)



8. Week 7 - Categorization

8.1. Key takeaways

8.1.1. What is a distributed system?

A distributed system is a group of computers working together so that, to the user, they appear like a single computer.

8.1.2. How can it be categorized?

There is no single universally applicable categorization. Distributed systems can be categorized by transparency, fault tolerance, scalability, consistency, data replication, data partitioning, and heterogeneity.

Examples are tightly vs. loosely coupled, homogeneous vs. heterogeneous, small-scale vs. large-scale, stateful vs. stateless, decentralized systems, client-server, peer-to-peer, hybrid, monolithic vs. microservices, and synchronous vs. asynchronous communication.

8.1.3. What are transparencies?

Transparencies are ways a distributed system hides its distributed nature from users.

Examples:
- Location transparency: users do not know the physical location
- Access transparency: users access resources in one uniform way
- Migration/relocation transparency: users do not notice resources moved
- Replication transparency: replicas appear as one resource
- Concurrent transparency: users do not notice other users
- Failure transparency: users do not notice recovery mechanisms
- Security transparency: users are minimally aware of security mechanisms

8.2. Distributed System Definition

A distributed system in its simplest definition is a group of computers working together as to appear as a single computer to the user

8.3. Distributed Systems Categorization

Classification: degree of transparency, fault tolerance, scalability, consistency (strong, eventual, and causal), data replication, data partitioning, heterogeneity.

There is no single universally applicable categorization of distributed systems

Classifications:
- Tightly coupled: Processing elements or nodes have access to a common memory
- Loosely coupled: Do not

Homogeneous System: All processors within the system have the same type
Heterogeneous System: Contains processors of different types

Small-scale system: WebApp + Database
Large-scale system: 2+ machines

Stateful: Maintains state between requests (e.g databases, session-based applications)
Stateless: System treats each request independently (e.g REST APIs behind a load balancer)

Decentralized: Distributed in a technical sense but not owned by one actor

More classifications:

- Based on architecture (Client-server, peer-to-peer, hybrid, software architecture)
- Based on deployment architecture (monolithic vs. microservices)
- Based on communication model (synchronous or asynchronous)

8.3.1. Controlled distributed systems vs. fully decentralized systems

- 1 responsible organization
- Low churn
- Examples:
- Amazon DynamoDB
- Client/server
- “Secure environment”
- High availability
- Can be homogeneous / heterogeneous
- Mechanisms that work well:
- Master nodes, central coordinator
- Consistent hashing (DynamoDB, Cassandra)
- Network is under control or client/server → no NAT issues
- Consistency
- Leader election (Zookeeper/Zab, Paxos, Raft)
- Replication principles
- More replicas: higher availability, higher reliability, higher performance, better scalability, but: requires maintaining consistency in replicas
- Transparency principles apply

- N responsible organizations
- High churn
- Examples:
- BitTorrent
- Blockchain
- “Hostile environment”
- Unpredictable availability
- Is heterogeneous
- Mechanisms that work well:
- Consistent hashing (DHTs)
- Flooding/broadcasting - Bitcoin
- NAT and direct connectivity huge problem
- Consistency
- Weak consistency: DHTs
- Nakamoto consensus (aka proof of work)
- Proof of stake – Leader election, PBFT protocols, Is Bitcoin eventually consistent? — Some argue no, some argue it has even stronger guarantees
- Replication principles apply to fully decentralized systems as well
- Transparency principles apply

8.4. CAP theorem

States that distributed data store cannot simultaneously be consistent, available and partition tolerant. Choose between consistency or availability.

- Consistency: Every node has the same consistent state
- Availability: Every non-failing node always returns a response
- Partition Tolerant: The system is designed to deal with broken communication between the nodes.

Example: Cassandra (NoSQL) is designed to be AP but can be CP.

8.5. Transparency in distributed systems

Distributed systems should hide its distributed nature:

- **Location transparency:** users should be unaware of the physical Location
- **Access transparency:** users should access resources in a single uniform way
- **Migration, relocation transparency:** users should not be aware that resources have moved
- **Replication transparency:** users should not be aware about replicas, it should appear as a single resource
- **Concurrent transparency:** user should not be aware of other users
- **Failure transparency:** user should not be aware of recovery mechanisms
- **Security transparency:** users should be minimally aware of security mechanisms

8.6. Fallacies of Distributed Computing

"Fallacies of Distributed Computing" means: Things developers often assume about networks, but should not assume.

1. Network is reliable
   - Even submarine cables can fail
2. Latency is zero
   - Ping to AUS is 300ms
3. Bandwidth is infinite
   - Send a bike courier with an 8TB disk, that arrives 10h later, or send the data with a 1Gibt/s link? 8 **1000** 8 / (10 **60** 60) = 1.7Gbit/s
4. The network is secure
   - Assume someone is listening
5. Topology doesn't change
   - Ping to AUS request can take different route than reply
6. There is one administrator
   - Sometimes your route goes from one company to another rival company
7. Transport cost is zero
   - Someone builds and maintains the network
8. The network is homogeneous
   - WiFi, cable, desktop, server, mobile

9. Week 8 - Communication Protocols

9.1. Key takeaways

9.1.1. How do network layers work?

Network layers provide services to the layer above them. Protocols allow an entity at one layer to interact with the same layer on another host.

Each PDU contains a protocol header and a payload, called the SDU.

9.1.2. What are the TCP mechanisms?

TCP is reliable and ordered. It uses retransmission, sequence numbers, ACKs, checksums, flow control, and congestion control.

Connections are established with SYN, SYN-ACK, ACK and terminated with FIN and ACK messages. Lost data can be detected with retransmission timeouts, duplicate ACKs, and selective ACKs.

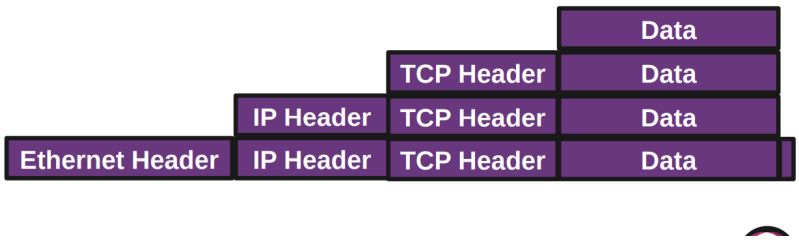
9.1.3. What are the problems of TCP, and how other protocols (HTTP/3) can improve that

TCP handshakes are not very flexible and can require multiple round trips, especially with TLS and DNS. TCP also requires packets to arrive in order, so one missing packet can delay everything.

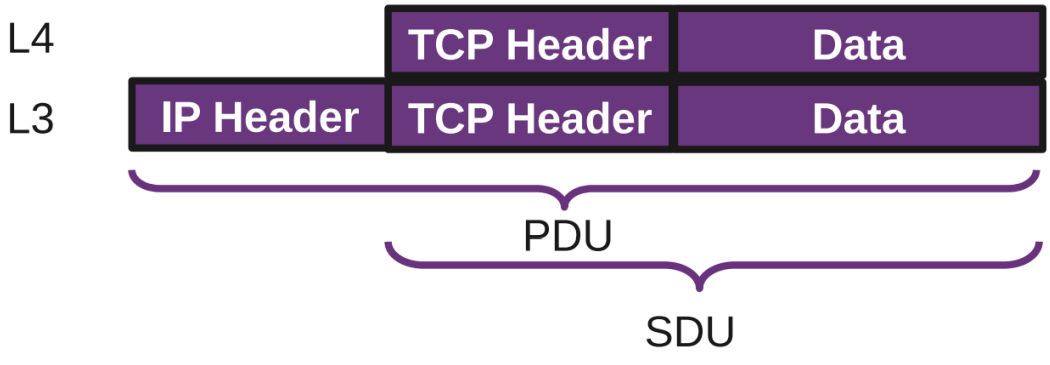
HTTP/3 uses QUIC, which has a 1 RTT connection and security handshake, 0 RTT for known connections, built-in security, and multiplexing where other streams can continue even if one stream has packet loss.

9.2. Networking Layers

OSI model	"Internet model"
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data link	Link
Pysical	



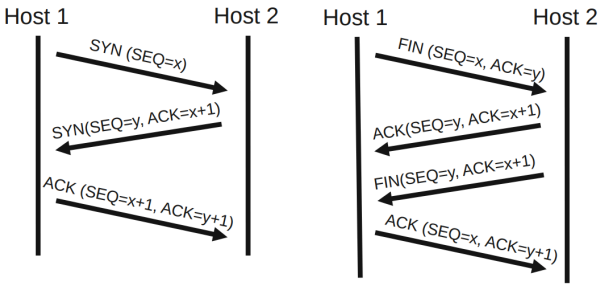
- Protocols enable an entity/instance to interact with an entity/instance at the same layer in another host
- Service definitions: provide functionality to an (N)-layer by an (N-1) layer
- Each PDU (protocol data unit) contains a protocol header and payload, the service data unit (SDU)



9.3. Layer 4 - Transport

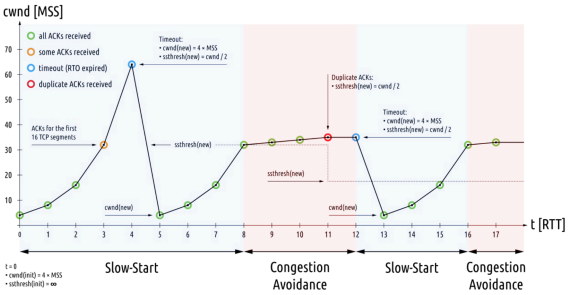
9.3.1. TCP

- Reliable (retransmission)
- Ordered
- Window - capacity of receiver
- Checksum - 16bit (crc16)
- TCP overhead: 20bytes (MTU 1500 bytes)
  - IP overhead: 20 bytes
  - Ethernet frame: 18 bytes (crc32)
- TCP tries to correct errors



- Connection establishment
  - SYN, SYN-ACK, ACK (three way)
  - Initiates TCP sessions: initial sequence number is random
- Connection termination
  - FIN, ACK + FIN, ACK (three/four way)
  - 3-way handshake, when host 1 sends a FIN and host 2 replies with a FIN & ACK
- Sequences and ACKs
  - Identification each byte of data
  - Order of the bytes -> reconstruction
  - Detecting lost data: RTO, DupACK
- Retransmission timeout
  - If no ACK is received after timeout (e.g 2xRTT), resend
- Duplicate cumulative acknowledgements, selective ACK
  - ACKs for last consecutive packets
  - 3 times same ACK -> retransmit missing packets (fast retransmit)

- Flow Control
  - Sender is not overwhelming a receiver
  - Back pressure
  - Sliding window:
    - Receiver specifies the amount of additionally received data in bytes that can be buffered
    - Sender up to that amount of data before the ACK
- Congestion Control
  - slow-start
  - congestion avoidance



9.3.2. TCP/IP from an Application Developer View

Problem: TCP handshake is not flexible

- You need a handshake (1RT)
  - If you want to make sure the other side accepts packets (and not drop it), ensure both sides are ready to transmit and receive data
  - If you want to exchange public / private keys
- TCP + Security = at least 2 RT

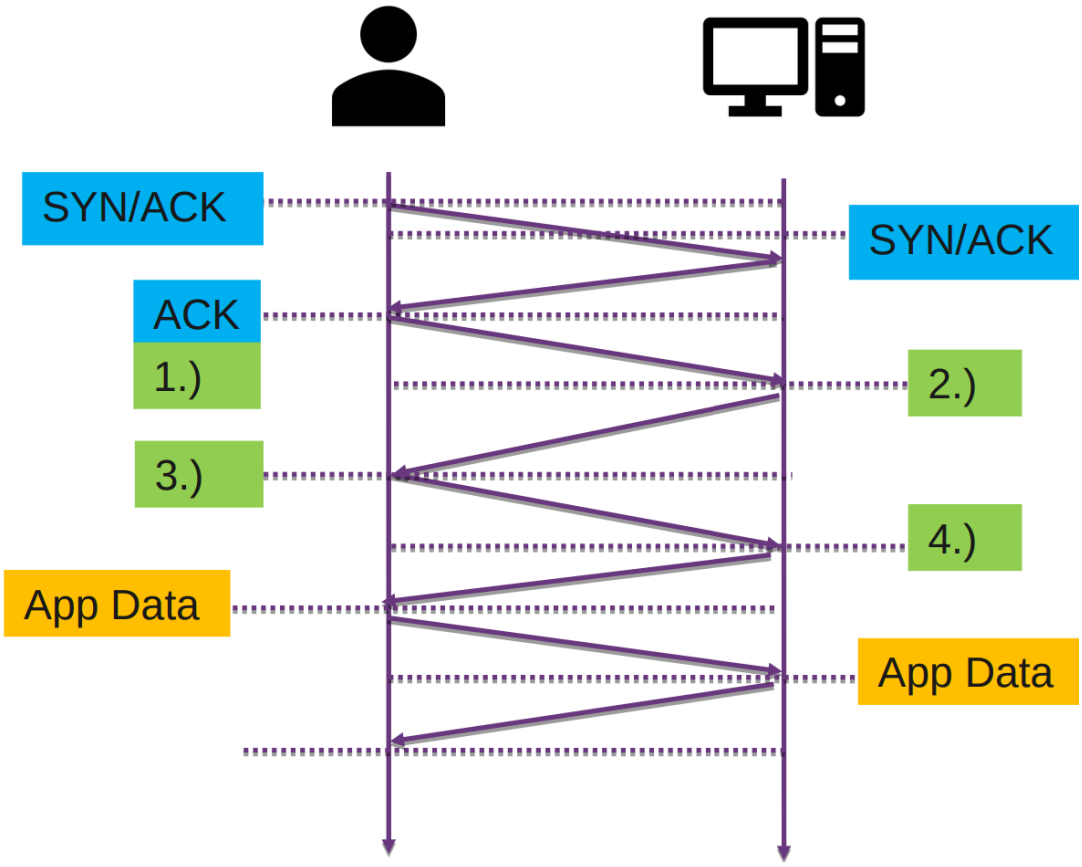
- DNS query may be required too: 3 RT
- Old security protocols add RT: 4 RT
- Worst case: 1.4s before data can be sent

9.3.3. TCP + TLS

Security: Transport Layer Security (TLS)

1. "client hello" lists cryptographic information, TLS version, ciphers/keys
2. "server hello" chosen cipher, the session ID, random bytes, digital certificate (checked by client), optional: "client certificate request"
3. Key exchange using random bytes, now server and client can calc secret Key
4. "finished" message, encrypted with the secret key

Total: 3 RTT to send first byte, 4 RTT to receive first byte



TLS 1.3: 1 RTT instead of 2

- Client Hello, Key Share
- Server Hello, key share, verify certificate, finished

9.3.4. QUIC / HTTP3

- QUIC: 1RTT connection + security handshake
- For known connections 0 RTT
- Built-in security
- HTTP/2 introduced multiplexing
- If one stream has packet loss, other streams can continue normally

The problem with HTTP/2: TCP requires packets to arrive in order, one missing packet can delay everything

Downsides of HTTP/3:
- NAT: TCP (SYN, ACK, FIN) knows when connection ends, with UDP you get many timeouts (creates many entries)
- HTTP header compression: Can reference previous headers which can create dependency
- Cannot benefit from TCP optimizations

9.3.5. UDP
- UDP is used for DNS, streaming audio and video
- Simple connectionless communication model
- No guarantee
  - Delivery
  - Ordering
  - Duplicate protection

9.3.6. Layer 4 - SCTP
- Message-based
- Allows data to be divided into multiple streams
- Syn cookies - SCTP uses a four-way handshake with a signed cookie
- Multi-homing multiple IP addresses of endpoints
- Not widely used
  - Problems with home routers
  - Used by WebRTC but tunneled over UDP

9.3.7. DDoS Amplification Attacks
DDoS amplification abuses UDP services that respond with more data than they receive. The attacker spoofs the victim’s address, so the server sends the large response to the victim. Tools like hping3 can create such spoofed UDP packets, while normal programming languages usually do not support this easily.

9.3.8. Comparison

TCP	UDP	SCTP	QUIC
Transport layer	Transport layer	Transport layer	Transport layer
Connection oriented	Connection less	Connection oriented	Connection oriented
Reliable transfer	Unreliable transfer	Reliable transfer	Reliable transfer
Streams	Messages	Messages	Multistream
Guaranteed order	Unordered	User can choose	Guaranteed order
Widely used - HTTP/1, HTTP/2	Widely used - DNS, HTTP/3	WebRTC	HTTP/3
Flow and congestion control	No flow, congestion	Flow and congestion control	Flow and congestion control
Heavyweight	Lightweight	Heavyweight	Heavyweight
Error checking and recovery	Error checking, no recovery	Error checking and recovery	Integrity check

Table 1: Comparison of transport protocols

9.4. Alternatives to QUIC

There are several alternative transport protocols or libraries, but most have drawbacks:

- KCP: A transport protocol without built-in encryption.
  - GFCP: A variant of KCP.
- UDT: No longer maintained.
- utp4j: Java implementation of Micro Transport Protocol, later handed over to Tribler at Delft University of Technology.
- Other related projects include:
  - 0-RTT protocol in Go
  - ATP, a peer-to-peer protocol
  - P2P library in Go

9.5. QOI - The quite ok image format

- Simple version of png
- Enocoder / decoder in 300 loc
- PNG generates smaller images, QOI is much faster
- Compression not much worse but simpler

10. Week 9 - Application Protocols

10.1. Key takeways

10.1.1. Overview over important protocols on layer 7

Layer 7 application protocols in this document include HTTP, WebSockets, Server Sent Events, WebRTC, DNS, Let's Encrypt / ACME, and mail protocols such as SMTP, POP, and IMAP.

10.1.2. Custom protocols, ASN.1, RPC, HTTP, JSON, WebSockets, Server Sent Events

Custom protocols can use less space, encode and decode faster, avoid unnecessary fields, and improve performance. Their downsides are development effort, testing/debugging, compatibility, and documentation.

ASN.1 is used for serialization/deserialization and certificates. RPC lets programs execute code on remote machines; examples include gRPC with HTTP/2 and Protocol Buffers. JSON is a human-readable data format, HTTP is the transport, and REST/RPC describe the API style.

WebSockets provide full-duplex communication over TCP after an HTTP upgrade. Server Sent Events are one-way communication from server to browser over a kept-open HTTP connection.

10.1.3. Bencoding, WebRTC, DNS, Let's Encrypt, Mail Protocols

WebRTC enables browser-to-browser real-time communication and uses STUN, TURN, and ICE for NAT traversal. DNS translates domain names to IP addresses and uses records such as SOA, NS, MX, A/AAAA, TXT, and PTR.

Let's Encrypt is a non-profit CA for automated TLS certificates via ACME. Mail protocols include SMTP for sending/receiving emails, POP, and IMAP. Mail security and spam prevention include STARTTLS, SMTPS, SPF, DKIM, DMARC, BIMI, greylisting, DNSBL, SURBL, and Bayesian analysis.

10.2. Protocols

Custom Protocols:
+ less space
+ faster encoding/decoding (custom)
+ fewer unecessary fields
+ better performance

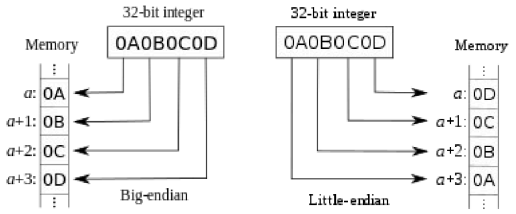
- development
- testing/debugging
- compatibility
- documentation

Protocol generators:
- Thrift: RPC framework from facebook, IDL and binary protocol
- Avro: data serialization system, remote procedure call and data serialization framework - Hadoop (Big-data framework)
- Protocol Buffers: data serialization from Google, IDL -> goals: smaller and faster than XML
- ASN.1: serializ. / deserializ, used for certs

IDL = Interface Description Language -> describe the protocol once, the tool generates code for different programming languages

Little-endian / big endian:

- Sequential order where bytes are converted into numbers
- Networking (TCP headers): Big-endian
- Most CPUs: Little-endian



10.3. RPC Examples

Remote Procedure Call: lets programs execute code on remote machines.

Examples:

- gRPC: Uses HTTP/2 for transport, uses Protocol Buffers
- Features: Authentication, bidirectional streaming and flow control, blocking or nonblocking bindings, and cancellation and timeouts

10.3.1. JSON/REST/HTTP

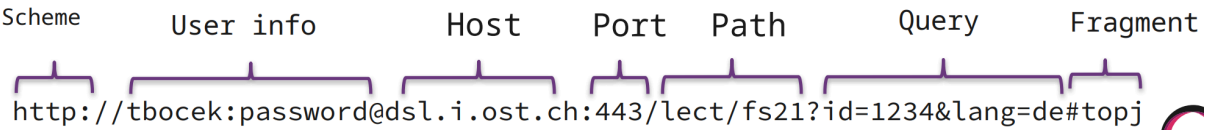
Human-readable text to transmit data. Can sometimes be RPC:

- REST ideally stateless, RPC can be stateful
- REST responses contain all metadata, RPC often needs interface definitions
- Parsing overhead: JSON slower than binary protocol
- Often used for web apps

So JSON is just the data format, HTTP is the transport, and REST/RPC describe the API style.

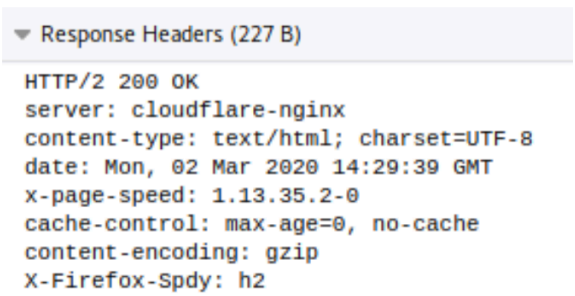
10.4. Application Protocol: HTTP

- Foundation of web data communication
- Request/response model, resources identified by URL
- URL structure: scheme, user info, host, port, path, query, fragment
- Stateless (server keeps no state)
- Browser sends additional headers (User-Agent, Accept, Accept-Encoding)



Response: header + status code + content body

Header:



Status Codes:

- 1xx (informational),
- 2xx (success – 200 OK),
- 3xx (redirection),
- 4xx (client error – 404 Not Found, 403 Forbidden),
- 5xx (server error)

Body/Content: <!DOCTYPE html> …

Request Methods: GET, HEAD, POST, PUT, DELETE, TRACE, OPTIONS, CONNECT, PATCH

10.5. WebSockets

- Full-duplex (two way) communication over TCP (REST/JSON is in one direction)
- HTTP handshake, then upgrade to communication channel (WebSocket)
- With SSL/TLS -> `wss://`
- Example usages: WhatsApp, Google Docs, YouTube
- Many languages support it, e.g Golang
- How can the server notify the browser
  - Polling:
    - Short: Request e.g every 0.5s
    - Long: request until timeout or reply
  - Server Sent Events (alternative) SSE
    - One way communication from server to browser
    - Server receives a regular HTTP request, keeps connection open (Accept or Content-Type: text/event-stream)
    - Native browser support via EventSource API
    - Automatic reconnection if connection is lost
    - Max number of concurrent connections per domain

10.6. WebRTC

- Browser to browser communication (P2P)
- Real time communication (RTC) via API
- Goal: eliminate plugins or native apps
- Supported by: Google, MS, Mozilla, Opera, Apple
- Replaced need for Flash, Java plugins -> allows e.g Chrome <-> Communication
- Used in WhatsApp, Facebook Messenger, MS Teams etc.

10.6.1. Concerns

- HTML browsers get bloated
  - Several GB RAM to open couple of tabs (Hint: adblocker)
- WebRTC API could be simplified
- Security concerns: May reveal IP information despite using VPN/tor
- WebRTC forbids unencrypted communication
  - DTLS (data), SRTP (media)
  - Complexity - SCTP over DTLS over UDP

10.6.2. WebRTC NAT Traversal

WebRTC allows browsers to communicate directly with each other, even if they are behind NATs or firewalls. The developer does not need to handle NAT directly, because WebRTC uses STUN, TURN and ICE as abstractions.

10.6.2.1. STUN

STUN is used to discover which public IP address and port a client has from the outside.

```
Client asks STUN server: Who am I?
STUN server replies: You are public-ip:port
```

STUN can also help detect the NAT type. However, STUN is not a complete NAT traversal solution by itself. It only discovers possible connection addresses, but some NATs or firewalls may still block direct connections.

10.6.2.2. TURN

TURN is used when a direct peer-to-peer connection is not possible. In this case, the traffic is relayed through a TURN server.

```
Client A <-> TURN server <-> Client B
```

This is more reliable, but less efficient because all data or media must pass through the relay server. TURN can use UDP, TCP or TLS, which is useful because some firewalls block UDP entirely.

10.6.2.3. ICE

ICE stands for Interactive Connectivity Establishment. It is the overall process that collects and tests possible connection paths between peers.

ICE candidates can include:

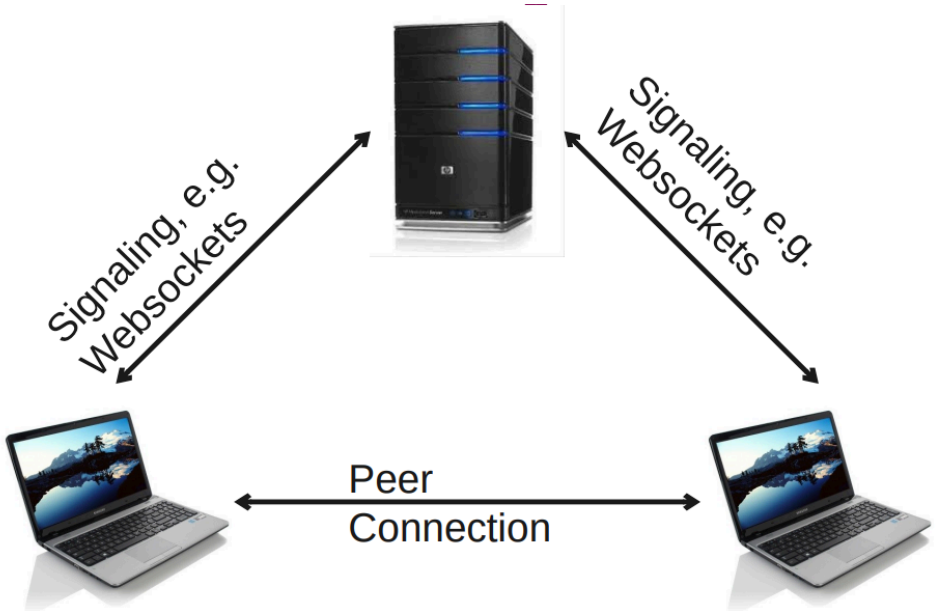
- local IP addresses
- public addresses discovered via STUN
- relay addresses from TURN

The peers exchange these candidates and test which connection works best.

1. Try direct connection
2. Try STUN-discovered public address
3. If that fails, use TURN relay

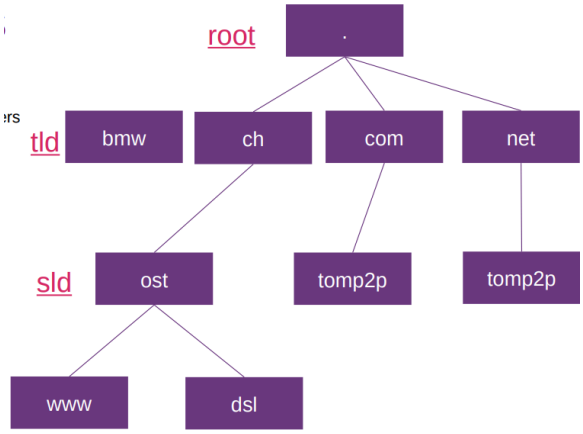
In short, STUN discovers addresses, TURN provides a relay fallback and ICE chooses the best working connection path.

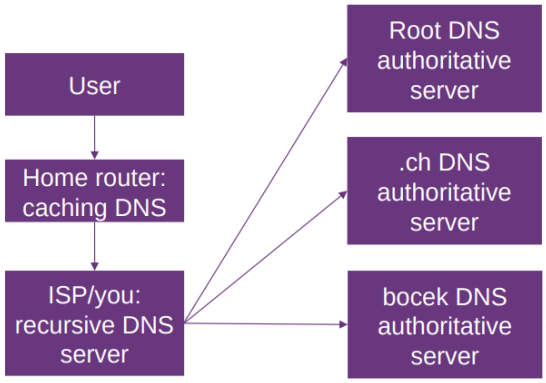
10.6.3. WebRTC Architecture



10.7. Application Protocol: DNS

Translates human-readable domain names to IP addresses “phone book of the internet”





- No special characters: ASCII
- Punycode: bücher.tld -> xn–bcher-kva.tld
- Hierarchical and decentralized naming system for computers
- UDP Port 53
- Primary + secondary DNS for redundancy
- Typical setup
  - User
  - Caching/forwarding DNS
  - Recursive servers: DNS name resolution for applications (e.g bind/unbound)
  - Authoritative servers: providing a definitive answer
    - Authoritative DNS: allows others to find your domain
    - Recursive DNS: allows you to resolve other domains
- Restriction to 13 root servers due to 512 byte packet limit
  - With anycast 2000 root server instances worldwide
  - l.root-servers.net , 1 root IP with anycast mirrored in 90 locations
  - Are regularly attacked (e.g DDoS)
  - Root zone managed by ICANN
- TTL defines the duration in seconds that the record may be cached by any resolver. “0” means no cache. Recommendation: > 1d

Type of records:

- SOA: Start of Authority
- NS: Name Server Record
- MX: Name and relative preference of mail servers
- A/AAAA: IPv4/IPv6 address record
- TXT: arbitrary and unformatted text
- PTR: opposite of A/AAAA

10.7.1. DNS Security

TSIG can be used to run a DynDNS service. It allows DNS updates to be authenticated before they are written to the DNS database. It uses a shared secret and cryptographic hashing.

DNSSEC is a DNS security extension. It provides authentication and data integrity, but not confidentiality. This means the receiver can verify that DNS data is authentic and unchanged, but the DNS query itself is still visible.

DNSSEC can also be used to bootstrap other security systems, for example certificates, SSH fingerprints or IPsec public keys.

Important DNSSEC keys:
- KSK: Key Signing Key, signs the ZSK public key
- ZSK: Zone Signing Key, signs the DNS records / RRsets

Important DNSSEC record types:
- RRSIG: signature for resource record sets
- DNSKEY: contains public DNSSEC keys
- DS: Delegation Signer record in the parent zone, used to link the trust chain

Using two keys makes key management easier. The ZSK can be changed more often, while the KSK is used to validate the ZSK.

10.7.1.1. DoT and DoH

DNSSEC provides signatures, while DoT and DoH provide encryption. Therefore, DNSSEC protects authenticity and integrity, while DoT and DoH protect confidentiality of DNS lookups in transit.

10.7.1.1.1. DoH: DNS over HTTPS

DoH sends DNS queries over HTTPS using HTTP/2 on port 443.

Advantages:
- DNS traffic looks like normal HTTPS traffic
- Harder to distinguish or block separately from web traffic
- Easy to deploy because DNS responses can be served like web pages
- Browsers can perform DNS queries directly using JavaScript

Disadvantages:
- Difficult client upgrade path because applications may need their own DoH support
- Performance includes TCP and TLS handshake cost, usually 2 to 3 RTT
- Can centralize DNS traffic at large providers

10.7.1.1.2. DoT: DNS over TLS

DoT sends DNS queries over TLS on the separate port 853.

Advantages:
- Provides confidentiality for DNS lookups in transit
- Easy for clients to test: try DoT on port 853 and fall back to normal DNS over port 53 if needed
- Widely supported by DNS software and public resolvers

Disadvantages:
- Easier to block because it uses a dedicated port
- Performance includes TCP and TLS handshake cost, usually 2 to 3 RTT

10.8. Let’s encrypt

Non-profit CA that provides certificates for TLS. **Domain-validation** certificates only. Automated renewal via ACME protocol.

- Certbot is an ACME client used to request and renew TLS certificates from Let’s Encrypt.
- With the webroot challenge, the challenge file must be placed where Let’s Encrypt can access it.
- The generated certificate and key paths are configured in Nginx.
- Renewal should be automated, for example with certbot renew and an Nginx reload.

- Caddy and Traefik already support ACME directly and can handle certificates automatically.

10.9. Mail protocols

Core protocols:

- SMTP: Simple Mail Transfer Protocol
- POP
- IMAP

10.9.1. SMTP

- Standard protocol for sending/receiving emails
- DNS MX records for mail server discovery
- Port numbers:
  - 25: non-encrypted communication
  - 587: Encrypted submission from email
  - 465 (deprecated): for SMTPS (SMTP over SSL)

Mail flow:

- MUA (Mail User Agent) ->
- MTA (Mail Transfer Agent) ->
- MDA (Mail Delivery Agent) ->
- Recipients mailbox

Routing: domain == local -> MDA; external forward to recipients MTA

STARTTLS (port 25/587):

- Upgrades plain text to TLS on same port
- Backward compatible, recommended standard
- Can fall back to unencrypted -> vulnerable to downgrade attacks

SMTPS (port 465):

- Guaranteed encryption from the beginning
- Port confusion due to deprecation/revival history

Greylisting:

- Temporarily rejects emails from unknown senders
  - First email from unknown sender -> temporary rejection (“try again later”)
  - Legitimate servers retry after delay (per SMTP standard)
  - On retry -> sender recognized, email accepted
  - Subsequent emails pass without delay
  - Reduces spam, low resource usage, simple to implement
  - Delays legitimate emails initially, spammers may adapt

10.9.2. Spam prevention

- SURBL (Spam URI Real-time blocklists): checks urls in emails against blacklists
- DNS Blocklists (DNSBL): lists of IP addresses known to send spam
  - e.g UCEPROTECT
  - Mail servers query DNSBLs in real-time to accept/reject
- Bayesian analysis: machine learning to classify emails by content
  - Training data: junk/not-junk emails
  - Word probability: “discount” -> likely spam, “meeting” -> likely not
  - e.g, thunderbird adaptive junk filter

SPF (Sender Policy Framework):
- Domain owner specifies allowed sending servers (DNS TXT record)
  - Receivers verify sender IP against SPF record -> prevents spoofing

DKIM (DomainKeys Identified Mail):
- Digital signature linked to domains DNS
- Verifies email integrity and sender authenticity

DMARC (Domain-based Message Authentication, Reporting and Conformance):
- Builds on SPF + DKIM, adds policy (reject/quarantine/allow) and reporting

BIMI (Brand Indicators for Message Identification):
- Publish logo (SVG) in DNS TXT record, requires VMC (verified mark certificate) in most cases

11. Week 11 - Consensus

11.1. Key takeways

11.1.1. Why does scaling lead to consensus problems?

Scaling means running multiple components or servers. Because components can be unreliable, replicas can end up in inconsistent states, so the system needs consensus to agree on one value, leader, block, data, or version.

11.1.2. Fault models: crash faults vs. Byzantine faults

A crash fault means a component fails or becomes unavailable. A Byzantine fault is an arbitrary fault where a node can do more than crash, for example lie to reach an advantage.

11.1.3. Consensus algorithms: Paxos, Raft

Paxos guarantees that nodes only ever choose a single value, but it does not guarantee progress if a majority of nodes are unavailable. It uses proposers, acceptors, and learners in a prepare/promise phase and an accept phase.

Raft uses followers, candidates, and one leader. Followers expect heartbeats from the leader; if the leader is unavailable, a new election starts. The system is only available when a leader has been elected and is alive.

11.1.4. Consistency in DHTs: vDHT

vDHT uses no locking and no timestamps. Every update creates a new version. `get()` waits until replica peers have the latest version, while `put()` first prepares data with a short TTL and only confirms it if all replica peers are OK.

With heavy churn, `get()` may return forks, so the API user may need to abort or resolve conflicts manually.

11.1.5. CRDTs: conflict-free data structures

CRDTs are an alternative to consensus protocols like Paxos or Raft. They do not need a leader or majority and can still work during network partitions, but only for suitable data structures.

Their operations must be commutative, associative, and idempotent, so replicas can eventually reach the same state without conflicts.

11.1.6. Tradeoffs between approaches

Paxos and Raft provide consensus but need a majority or leader to make progress. vDHT avoids locking and timestamps, but conflicts or forks may need to be resolved by the application. CRDTs can work without a leader or majority, but are limited to suitable data structures and merge algorithms.

11.1.7. Different ways to deploy your service — High-level overview

Modern deployment is based on containers, but still needs a strategy. Deployment approaches include rolling deployment, blue-green deployment, canary releases, feature toggles, and big bang deployment.

Practical options include plain Docker Compose or Podman, Docker Swarm, Kubernetes for larger deployments, Ansible/Chef/Puppet for automation, hosted services like Railway, and Terraform for Infrastructure as Code.

11.2. Consensus

Definition: Consensus decision-making is a group decision-making process in which group members develop, and agree to support a decision in the best interest of the whole. Consensus defines leader (creates blocks, adds data, creates version).

Byzantine Fault: Arbitrary fault that occurs during the execution of an algorithm by a distributed system (not only crash but lie to reach an advantage)

Find consensus: Paxos, Raft, vDHT, Zookeeper

11.2.1. Paxos

Paxos is a consensus algorithm by Leslie Lamport that became important for fault-tolerant distributed systems. It was first rejected, later accepted and helped Lamport win the Turing Award. It is largely considered difficult to understand.

Problem: Due to unreliable components, run multiple components, i.e multiple servers leads to inconsistent state with replica

Paxos guarantees that nodes will only ever choose a single value, but does not guarantee that a value will be chosen if a majority of nodes are unavailable.

Roles: proposer, acceptor and learner

- Proposer proposes a value that it wants agreement upon
- Acceptor gets proposal, makes promises, sends result to learners
- Learners: majority of accepots must choose the same value

2 phases: prepare/promise phase, accept phase

Example: Proposer: prepare and accept requests, proposal number and value (n, v)

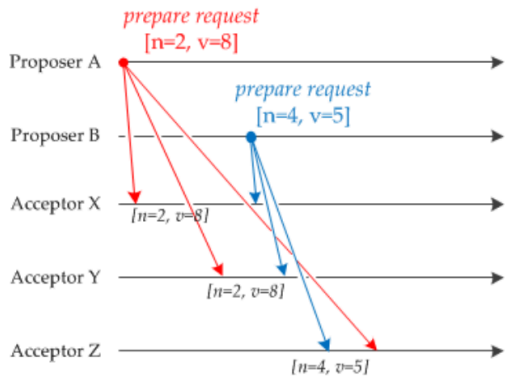
- Acceptor: already seen higher proposal number: ignore
- Acceptor: seen lower proposal number: send back highest accepted n,v

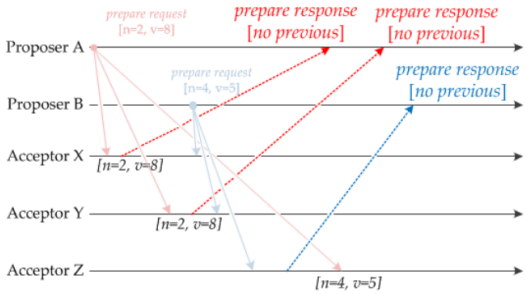
Proposer A and proposer B

- Acceptor Z receives B before A

If acceptor receives prepare request for the first time:

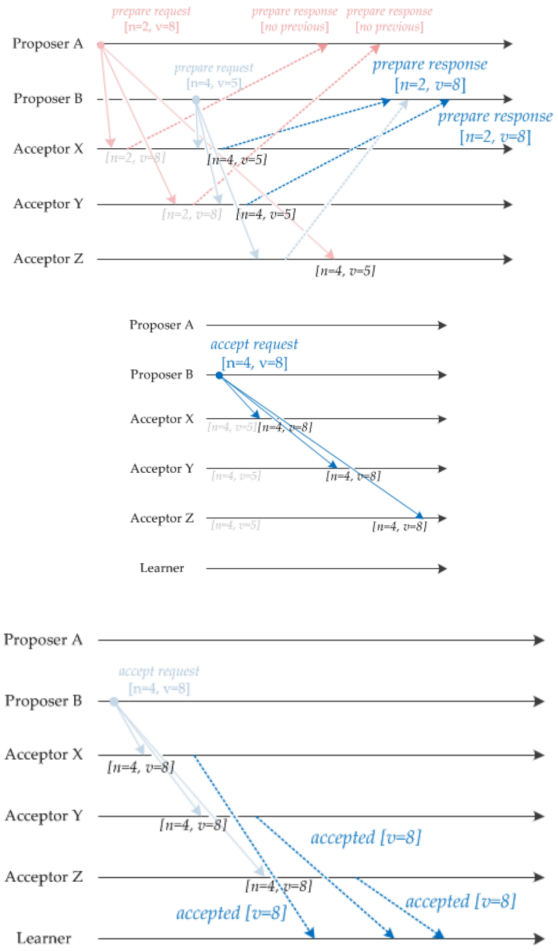
- Acceptor responds with a prepare response
- Promises never to accept another proposal with a lower proposal number
- Acceptor Z receives proposer A's request, and acceptors X and Y receive proposer B's request.
  - Only accept requests with higher number, Acceptor Z not sending response (or negative response) to B.
- Proposer B sends an accept request to each acceptor containing the proposal number it previously used (n=4) and the value associated





with the highest proposal number among the prepare response messages it received (v=8)

- Proposer A sends its accept request before proposer B, but acceptor ignores them
- If an acceptor receives an accept request for a higher or equal proposal number than it has already seen, it accepts and sends a notification to every learner node.
  - A value is chosen by the Paxos algorithm when a learner discovers that a majority of acceptors have accepted a value
  - Once a value is chosen by Paxos, communication with other proposers cannot change value
  - If another proposer, sends a higher proposal number than has previously been seen, with a different value (e.g., n=6, v=7), each acceptor responds with the previous highest proposal (n=4, v=8)
  - This requires the proposer to send an accept request containing [n=6, v=8], which confirms the value that has already been chosen



11.2.1.1. Raft (multi paxos)

RAFT: Reliable, Replicated, Redundant, And Fault-Tolerant

- Follower, Candidate, leader
  - Raft implements leadership election
  - Once a leader has been elected, all decision-making within the protocol will then be driven only by the leader
  - Only one leader can exist at a single time
- Each follower has a timeout (typically between 150 and 300ms) in which it expects the heartbeat from the leader
  - The system is only available when a leader has been elected and is alive
  - Otherwise, a new leader will be elected, and the system will remain unavailable for the duration of the vote
  - Starts election by increasing term counter, voting for itself, and sending a message to all other servers requesting their vote
  - If higher term is received become follower, if not, leader

11.3. Consistency

vDHT Basics:

- No locking, no timestamps
- Every update - new version

1. Get() latest version, check if all replica peers have the latest version, if not wait and try again
   - May add delay
   - Wait until update is completed
2. put() prepared with data and short TTL, if status is OK on all replica peers, go ahead, otherwise, remove the data and go to step 1
   - Data can be either send now or with the confirm, if sent now we are optimistic
   - Peer marks the value as prepared, other put() fail on that key. If nothing happens, TTL
   - Value linked to previous version(s) (hash)
3. put() confirmed, don't send the data, just remove the prepared flag and reset TTL

- In case of heavy churn (constant joining and leavig), API user needs to resolve
  - Get latest version may return fork
  - Abort or resolve (join) manually

11.4. CRDT

CRDTs are an alternative to consensus protocols like Paxos or Raft. They do not need a leader or a majority, so they can still work during network partitions. However, they are limited to suitable data structures.

A vDHT is similar to an operation-based CRDT, but conflict resolution is delegated to the application. This means forks or conflicting versions may need to be resolved manually.

CRDT stands for Conflict-free Replicated Data Type. It can be imagined like Git, but without merge conflicts.

For CRDTs to work correctly, their operations must be:

- Commutative: $x \bullet y = y \bullet x$
- Associative: $(x \bullet y) \bullet z = x \bullet (y \bullet z)$
- Idempotent: $x \bullet x = x$

11.4.1. Example: G-Counter

A G-Counter stores one counter value per machine.

```
A:6 B:0 C:0
A:0 B:3 C:0
A:0 B:0 C:9
```

When replicas are merged, the maximum value of each counter position is kept:

```
merge(A:6 B:2 C:9, A:5 B:3 C:2) = A:6 B:3 C:9
```

Because this merge is commutative, associative and idempotent, all replicas eventually reach the same state.

11.4.2. Collaborative Text Editing

A common CRDT example is collaborative text editing. Google Docs uses operational transformation, which usually requires a central server.

CRDTs can also work without a central server, for example in a peer-to-peer system. For this, suitable merge algorithms are needed.

One idea is to assign each character a position value between 0 and 1:

```
H    e    l    o
0.2  0.4  0.6  0.8
```

Concurrent edits can then insert characters between existing positions:

```
Insert "l" at 0.7
Insert "!" at 0.9
```

After merging:

```
H    e    l    l    o    !
0.2  0.4  0.6  0.7  0.8  0.9
```

This allows merging without direct conflicts. However, collisions can happen if two edits choose the same position, so the algorithm needs a strategy for this.

Another problem is interleaving: if two users insert multiple characters at the same place, the merged result may mix them in an unwanted order. Existing CRDT algorithms handle such cases.

Examples of CRDTs:

- G-Counter
- PN-Counter
- G-Set
- 2P-Set
- Yjs, a CRDT library

12. Week 11 - Deployment

Old deployments were manual: install packages, configure software, run scripts. This caused version problems, dependency conflicts, config drift, crashes, and scaling issues.

Containers help by packaging the app with its libraries, making it easier to move, scale, and run consistently. But deploying containers still needs a strategy.

12.1. Deployment Strategies

Rolling Deployment:

- New version is gradually deployed to replace the old version (doesn't take the entire system down at once)
  - Minimal downtime, low risk
  - Complexer, longer deployment times

Blue-Green Deployment:

- 2 environments
  - Blue: Current prod
  - Green: Current prod with new release
- Test, then switch
- Instant rollback, 0 downtime
- Issue: 2 prod environments, keep data in sync

Canary Releases:

- New version to a small group of users or server first, if all goes well, more users
- Reduces risks, gives user feedback
- Complex, leads to inconsistencies

Feature toggle:

- Fine grained canary, set feature for specific users
- More risk reduction, specific user feedback

- Increases complexity of codebase and config management

Big bang:
- Deploy everything at once
  - Simple
  - High risk, limited rollback

12.2. Practical Deployment
- Containerization is the basis for modern deployment.
- Ansible, Chef and Puppet are used for deployment automation.
  - Ansible uses playbooks and an SSH host list.
  - The target hosts should usually run the same OS/package manager, for example apt or yum.
- Docker Swarm works with docker-compose.yml.
  - It lets you package and deploy the application in the same way across platforms.
  - It is simpler than Kubernetes.
- Kubernetes is widely used, especially for larger deployments.
- Plain Docker Compose or Podman is the simplest option.
  - Example: run docker compose up -d –build over SSH.
- Ansible:
  - Does not require agents on the servers.
  - Is push-based, so commands are pushed from your machine/control node.
  - Uses SSH to connect to hosts.
  - Runs with commands like ansible-playbook playbook.yml.
- A more basic alternative is pssh/ssh, but it is less structured than Ansible.

12.2.1. Podman / Docker Swarm
Podman:
- Daemonless
- Simpler but deployment needs more works
- Quadlet:
  - Run container under systemd in a declarative way
  - Tools to convert podman/compose
  - Auto-update from registry
- Many variations, tools, helpers: podman-compose

Docker Swarm:
- Deploy with docker-compose.yml
- Built into docker
  - docker swarm - manage swarm
  - docker stack - manage deployments
  - docker node - manage nodes
- Scheduler is responsible for placement of containers to nodes

12.2.2. Kubernetes (K8s)
- Container orchestration: deployment, scaling, management
- Industry standard for complex applications

Why:
- High availability, fault tolerance (containers/nodes can crash)
- Auto-scaling based on demand

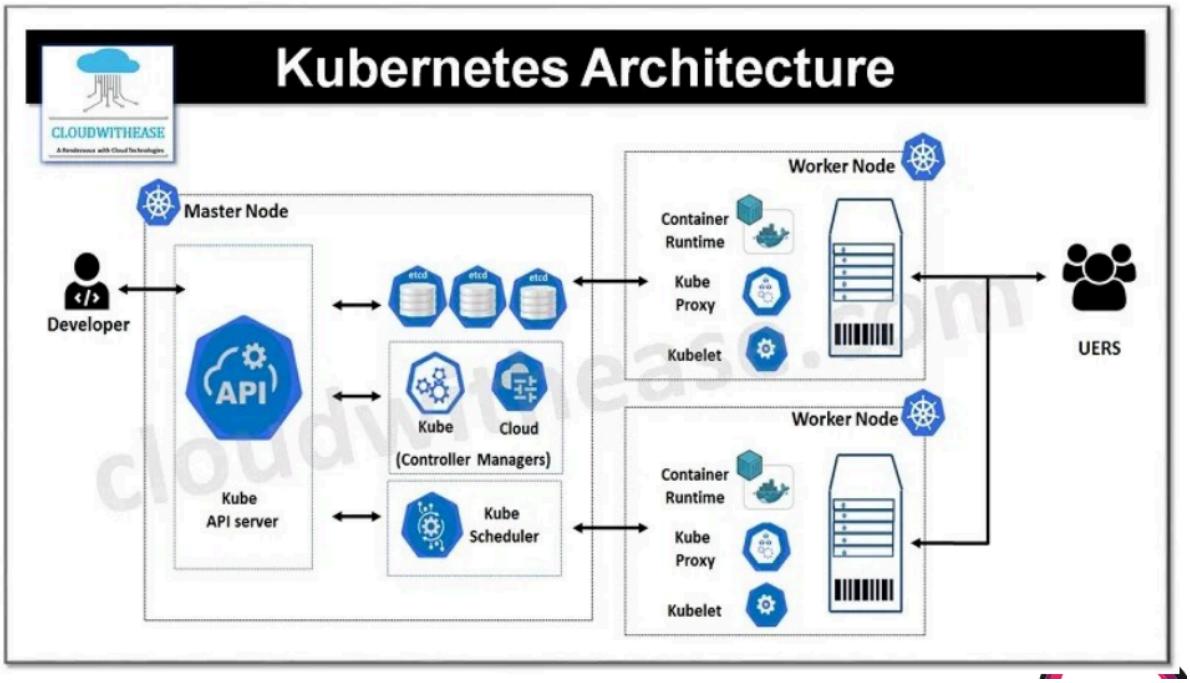
- Rolling updates and rollbacks built-in
- Ecosystem: Helm (package manager), operators

Design principles:
- Declarative config (YAML / JSON) - describe desired state
- Immutable containers - don’t store state, restart on health check fail
- Rollbacks possible but tricky with schema changes

Architecture: Master + Worker nodes
- Master: API server, etcd, controller manager, Scheduler
- Worker: kubelet, kube-proxy, container runtime (containerd, CRI-O)

Key concepts:
- Pod: smallest deployable unit, one or more containers
- Service: stable network endpoint for a set of pods
- Deployment: manages desired state, scale, HW limits
- ConfigMap: non-sensitive configuration data
- Secret: sensitive data (passwords, API keys), encrypted in etcd
- Volume: persistent storage, decoupled from prod lifecycle
- Namespace: segment cluster, multiple projects/teams isolated

Getting started:
- Local development: minikube, k3s
- kubectl: command-line tool
- GUI: kubernetes dashboard, lens



12.2.3. Services, Terraform

Services:
- AWS, App Runner
- Digital Ocean, App Platform
- Railway

- Git based deployment
- Zero config and HTTPS support
- Integrated Dbs
- Pull request -> preview
- Logs / monitoring

Terraform: Terraform is used for Infrastructure as Code.

That means you describe your infrastructure in configuration files instead of creating everything manually in a web dashboard.

For example, you can describe:
- servers
- databases
- networks
- load balancers
- DNS records
- permissions

Deployment Best Practices:
- Automate the deployment as much as possible
- Infrastructure as Code
- Immutable Infrastructure
  - Instead of SSH-ing into a server to update manually, you build new image and deploy that
- Health Checks / Monitoring
- Centralized Logging
- Rollback

13. Week 12

13.1. Key takeaways

13.1.1. Understand how Bitcoin works as a peer-to-peer system

Bitcoin has no central bank, transactions are created by peers and broadcast to the whole network.

13.1.2. Get familiar with UTXO, mining, and proof-of-work

Coins are spent as whole UTXO pieces, while miners create valid blocks by solving proof-of-work puzzles.

13.1.3. Be aware of risks like the 51% attack

An attacker with more than 50% computing power could influence transaction ordering and exclude transactions.

13.1.4. Reflect on advantages and disadvantages of Bitcoin

Bitcoin is decentralized and can have low fees and strong anonymity, but has high power use, volatility and scalability limits.

13.2. Introduction

- Bitcoin is an experimental digital currency.
- It is fully peer-2-peer, with no central entity like a central bank.
- It was first issued on Jan 3, 2009.
- The smallest unit is 0.00000001 BTC, called 1 satoshi.
- The initiator is unknown.
- Bitcoin can be exchanged for real currencies.
- There is a maximum of ~21 million BTC.
- Every transaction is broadcast to all peers, so they all know every transaction (~736 GByte).
- Transactions are validated by proof-of-work, a partial hash collision that is difficult to fake.
- Bitcoin does not rely on trust, but on strong cryptography.
- Bitcoin has weak anonymity.
  - Anonymity can be weakened through clustering. For example, if a transaction has multiple input addresses, those addresses are assumed to belong to the same wallet.
- BIP: Bitcoin Improvement Proposals

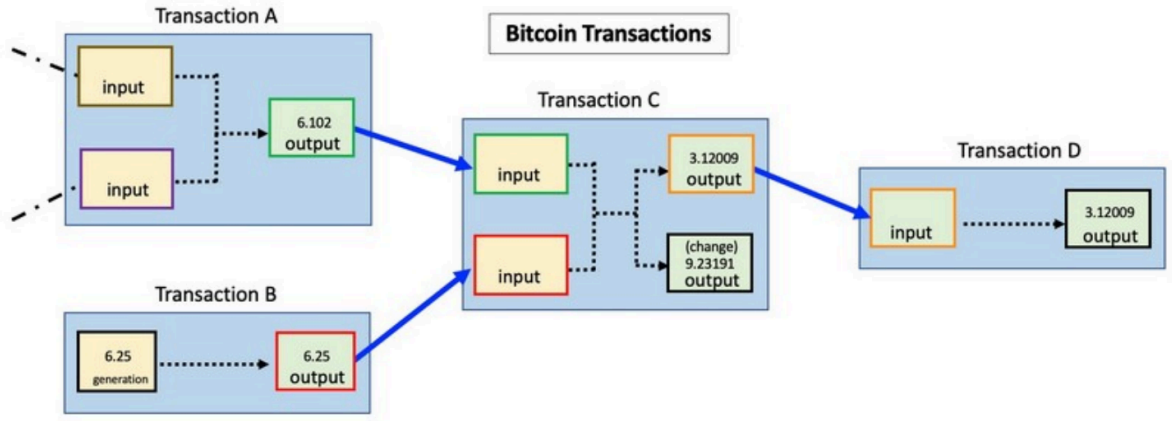
13.3. Mechanism

A wallet has public-private keys (wallet.dat)

- Public key, ECDSA 256 bit → Bitcoin address (can receive bitcoins)
- Simple address base58(RIPEM160(Sha256(ecdsa public key)))
  - E.g. 1GCeaKuhDYnNLNR6LGmBtKhPqEJD4KeEtF
- Private key used for signing transactions

13.3.1. Transaction

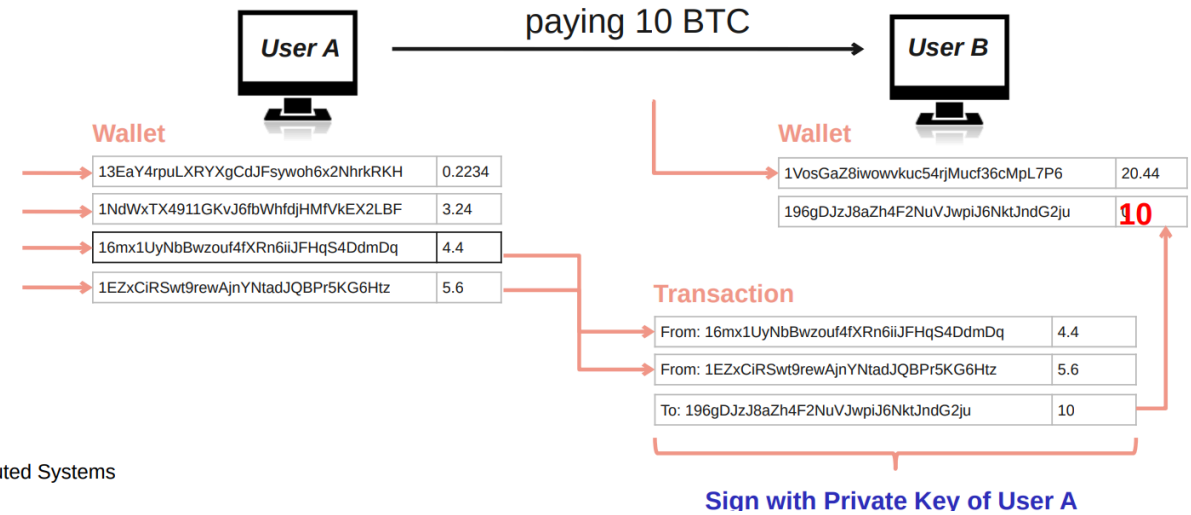
- Peer A wants to send BTC to peer B → creates transaction message
- Transaction contains input / output
  - where the BTC came from and where it goes
- Peer A broadcasts the transaction to all the peers in the network
- Transaction stored in blocks → block is created / verified 10min



13.4. Key Bitcoin Operation

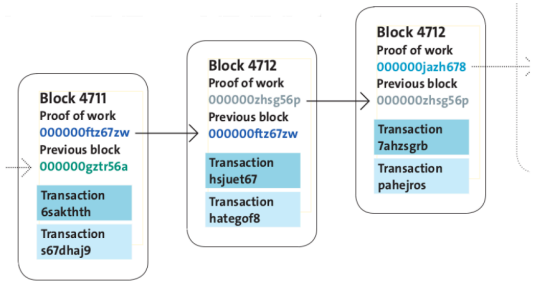
Private key authorizes the transation:

- Stolen keys can be used by thieves
- Key lost = coins lost
- In UTXO (unspent transaction output) systems, coins are stored as unspent pieces from earlier transactions
- When one piece is used, the whole piece is spent, the rest comes back as new change



13.5. Blockchain

- Transactions are collected in blocks
  - New block created approximately every 10 min
- Blocks contain solved crypto puzzles
  - In the form of partial hash collisions (SHA256)
  - Miners search for a block hash with a required pattern, e.g. many leading zeroes
  - More required zeroes = higher difficulty
- A block has a pointer to previous block →
  - This creates the blockchain



  - Changing an old block changes its hash, so all following blocks would need to be recalculated
- Creation of blocks is called mining (reward)
  - Mining / creating blocks → Miner get currently 3.125 BTC per creation
    - adjustable difficulty 6 blocks / h
    - Sometime in 2028 reward will be 1.5625

13.6. Mining mechanism

There are a couple of big miners (e.g the 1 million bitcoins of satoshi whose whereabouts are unknown).

- Mining = creating valid blocks
- Blocks are linked to previous blocks
  - Longest block survive (most difficult)
- Different level of confirmations
  - 3-6 block conf. is considered secure
- Dangerous if someone has more than 50% computing power
  - Can exclude and modify the ordering of transactions

Mining evolution:

1. CPU
2. GPU
3. FPGA (field programmable gate array)
4. ASIC Farms
   1. Application-Specific Integrated Circuit
   2. Whole datacenter

ASIC scenario:

- Old miner with 70 GHash/s
- Generates `~0.004 CHF` per day in 2026
- Uses 700W
  - 16.8kWh
  - Cost per day 3.69 CHF -> (Hochtarif, 0.2CHF per kWh)
  - Cost per day 2.18 CHF -> (Niedertarif, 0.15CHF per kWh)

13.7. Discussion

Advantages:

- Low tx fees
- Scalable (hardware/storage gets faster)
- Anonymity (preserving privacy)
- No major crashes
- Decentralized
- Other blockchain use cases

Disadvantages:

- High power consumption
- Not scalable (less transactions per sec than visa)
- Anonymity (illegal activities)
- Volatile exchange rates
- Central elements (devs)